Dauphine | PSL
UNIVERSITÉ PARIS

Master M2 111 - Économie internationale et développement

Climate Change and Sovereign Debt Sustainability

Author
Alexandre Saint

Supervisors
Anne-Laure Delatte
Loïc Henry

Master's thesis • Academic year 2025 – 2026

Abstract

We analyze the link between climate change and debt sustainability in France within a dynamic general equilibrium (DGE) model, in order to simulate the economic dynamics resulting from the shock represented by the net zero-transition. We replicate the model of Seghini and Dees (2025), a DGE that includes the green transition, climate damages and a fiscal block, and add the damage function found in Bilal and Känzig (2026), on the upper end of the damage spectrum found in the literature. We find that in the case of ambitious mitigation objectives, relying solely on public money to finance the transition can prove risky, due to the difference between public and private mitigation cost structures. The physical damages produced by climate change, in addition to reducing GDP, exacerbate the fiscal stress that may arise from government-funded transition policies, with the peak annual default probability rising from about 12% to above 20% in the worst case scenario. The cost to output, alongside consumption and investment, rises greatly under 3°C, with losses of about a third of output under the most severe damage specification, that of Bilal & Känzig.

Statement on the use of artificial intelligence

The research presented in this thesis, including the modelling choices, the computational implementation, the analysis of the results and the entirety of the written text, is my own work. Artificial intelligence assistants (Claude and Claude Code, Anthropic) were used exclusively as technical support, for four purposes: help in understanding some concepts and mathematical derivations, document formatting and typesetting, the optimisation of the MATLAB and LaTeX code, and the facilitation of mind-mapping during the exploratory phase of the project. No part of the text of this thesis was generated by artificial intelligence.

Contents

Abstract i

Statement on the use of AI ii

1 Introduction 1

2 Literature review 5

3 Model 8
- 3.1 Hypotheses and logic of the model . . . 8
- 3.2 Equations . . . 8
- 3.3 Implementation . . . 14
- 3.4 Adding a new damage function . . . 17

4 Results 23
- 4.1 Simulation results . . . 23
- 4.2 Policy implications . . . 30

5 Discussion 32

6 Conclusion 34

References 36

List of Figures 39

Appendices 40

A Mathematical derivation of the Bilal & Känzig damage function 40
- A.1 From the convolution integral to two differential equations . . . 40
- A.2 Exact discretisation at a quarterly frequency . . . 41
- A.3 Long run, initial condition and normalisation . . . 42

B Detrended and steady-state equations 43
- B.1 Detrending . . . 43
- B.2 Detrended equations . . . 44
- B.3 The two steady states . . . 46

1 Introduction

The summer of 2026 has been a sadly impressive reminder of the consequences of climate change. In France, it has beaten 2025's record of hottest summer ever registered, and it is likely to be beaten next year. The number and scale of fires witnessed in the country was unprecedented, and gives increasingly little doubt on the root cause of all this. One could pretext that such extreme events have already happened in the past, or that they are due to exceptional circumstances, but science is unanimous, and the facts speak for themselves : the rise in global temperatures observed over the past century, a product of the emission of greenhouse gases (GHG) resulting from economic activity, has coincided with a similar rise in the frequency and intensity of natural disasters. The world has indeed reached in 2024 an average mean temperature of 1.53°C above pre-industrial averages, or 1.28°C if we look at the 10-year average, this last number being actually more accurate since the trend is more informative than the fluctuations around it (Figure 1). Fifty years ago, this figure was only around 0.5°C, and since then, the number of extreme events per year, as registered by EM-DAT, has risen from less than 100 to more than 400, of which wildfires actually occupy only a tiny fraction, the vast majority being represented by floods and storms.

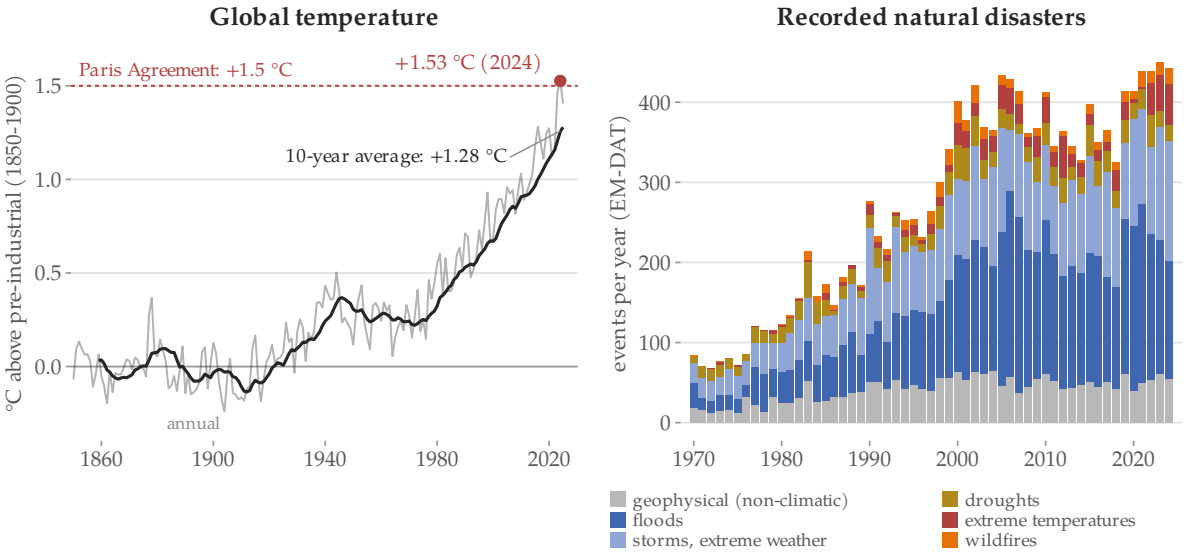


Figure 1: Global warming and recorded natural disasters
Sources: HadCRUT5 (Met Office) and EM-DAT (CRED, UCLouvain), via Our World in Data.

These climate events hurt the economy by reducing output, mainly via two channels : capital destruction and damages to productivity. Storms and floods may for instance impair or destroy buildings, factories, machines and extreme temperatures, say, may degrade working conditions and impact workers' health, resulting in less value produced in the same amount of time. To prevent those events from happening, it is necessary to de-carbonize the economy, that is to say to reduce the CO2 emissions

produced by output, and make the "green transition". As Pisani-Ferry and Mahfouz (2023) point out, though technological progress plays a role via the decoupling output from emissions, it will not be enough to reach the goals set to maintain the planet in acceptable conditions, under +2°C from pre-industrial times as set by the Paris agreement. Some part of sobriety, or reduction in output per capita, will also be inevitable. The policies carried out to achieve this goal may also affect the economy by themselves so it is crucial to separate their effects from those of climate change itself. Those risks, often respectively referred to as transition risk and physical risk, will be extensively studied in this work. Another key element in dealing with the economic consequences of climate change is adaptation. As we are already experiencing those consequences and a rise in temperature is inevitable, it is also necessary to invest in infrastructure that make their impact manageable. This part won't be studied here, although an analysis can be found in Seghini et al. (2026).



Figure 2: Public debt, fiscal balance and interest-growth differential in France, Germany and Italy, 1980–2025

Sources: IMF, World Economic Outlook, April 2026 (debt, balance); Eurostat and World Bank (r-g).

How can this transition be made possible ? Whereas some companies have pledged to reach net-zero goals, the private sector as a whole may lack the incentives and long-term vision to carry the transition all by itself. Furthermore, a large part of the

effort relies on strategic investments − think of infrastructure, energy, transportation − to be made by the state. Public money is thus, with no surprise, expected to play a major role in the fight against climate change. However, especially in the case of France, the country studied in the present work, fiscal space happens to have been considerably reduced over the past decade. France's fiscal balance has never been in surplus since 1980, and its public debt reaches 116%, almost twice the threshold of the Maastricht criteria, the reference of EU treaties (Figure 2). Although this figure may not be as high as that of its Italian neighbor, France's public debt to GDP ratio has been following a strictly upward path from the 21% level it displayed in 1980. A key moment in this trajectory is the 2008-2011 period, up to which France and Germany followed a similar path, but after which the two countries started to diverge, Germany having managed to stabilize around 60%. One has to note that this four-year period is nonetheless quite peculiar as it covers two major crises, the subprime crisis and the euro area debt crisis, and that the "divergence" seen in the graphs reflects two different ways of managing it. In addition, the situation in which France could easily borrow at low cost is now over. As the bottom panels of Figure 2 show, the interest-growth differential turned back positive after almost a decade in negative territory, which means that GDP growth no longer absorbs the debt, and that debt now feeds on itself. At a time when every spending budget seems a priority, some may discard investments in the green transition as "not as necessary as we thought", or "something to be done later on". This would be a disastrous mistake.

Indeed, the money saved on mitigation today is a small fraction of what it will cost in output tomorrow. Over the long run the cost of inaction largely outweighs the cost of action, a conclusion on which the literature is unusually consensual. Stern (2006) had already concluded that unmitigated climate change would cost at least the equivalent of a permanent 5 % loss of world consumption. Burke et al. (2015) later estimated that unmitigated warming would reduce average global incomes by roughly 23 % by 2100, and Bilal and Känzig (2026) that a warming path reaching 3 °C by 2100 would lower world GDP per capita by 53 % relative to a world without further warming.

Climate change and debt sustainability are thus profoundly related. As Agarwala et al. (2021) pointed out, both transition policies and climate damages impact the determinants of the public debt dynamics equation (1). The damages to output do not only damage growth g_t but also the primary surplus s_t (the fiscal balance before interest payments) through the reduction of tax revenues. Extreme events may also necessitate emergency spending by the state, such as medical expenses or the restoring of damaged public infrastructure. Finally transition policies relying on government expenses may also weigh on fiscal balances. The behavior of surpluses and public debt may consequently affect interest rates r_t, that may in turn feed back into debt

dynamics.

$$\Delta d_t = \frac{r_t - g_t}{1 + g_t} d_{t-1} - s_t \tag{1}$$

In this work, some of those channels are discarded, such as direct public spending entailed by extreme climate events. Capital destruction is also not taken into account. All physical damages resulting from climate change are captured by a "damage function" that negatively affects total factor productivity (see Section 3.4).

Although those relationships could be as well analyzed empirically, we chose to resort to a dynamic general equilibrium (DGE) model in order to have a comprehensive view of all the interrelations among the involved variables. Such a tool also makes it easy to simulate different scenarios by changing parameter values. We thus replicated the model of Seghini and Dees (2025), a DGE that includes the green transition, climate damages and a fiscal block in order to analyze the interrelations mentioned above. We add the damage function found in Bilal and Känzig (2026), on the upper end of the damage spectrum found in the literature, in order to see its impact on the dynamics of the model.

We find that in the case of ambitious mitigation objectives, compatible with global mean temperature staying under +1.5°C compared to pre-industrial times, relying solely on public money to finance the transition can prove risky, due to the difference between public and private mitigation cost structures. The physical damages produced by climate change, in addition to reducing GDP, exacerbate the negative effects of transition policies, with peak default probabilities rising above 20% in the worst case scenario. The cost to output, alongside consumption and investment, rises greatly under 3°C, with losses up to 50% in the most severe damage specification.

In the following chapters, we carry out a short literature review (Section 2), before presenting the model and its extension (Section 3). After laying out the results of our simulations and their implications (Section 4), we take a step back to reflect on our approach and areas of future research (Section 5) and conclude (Section 6).

2 Literature review

The question this work asks sits at the conjunction of two well established topics in the scientific literature : public debt sustainability and the economic consequences of climate change. However, only a few recent papers actually combine both aspects. A literature review has already been carried out by Seghini (2024), upon which the present section thus largely draws.

The study of the effects of climate change on the economy can be dated back to the Meadows report (Meadows et al., 1972), which predicted that our economic system, by promoting unlimited growth in a world with finite natural resources would eventually collapse. Later, Kaya (1990) gave the decomposition, shown in (2), of emissions (E) into population (P), income per head (Y/P), energy intensity (F/Y) of output and carbon intensity of energy (E/F) that still organises mitigation scenarios.

$$E = P \times \frac{Y}{P} \times \frac{F}{Y} \times \frac{E}{F} \tag{2}$$

This literature has experienced a series of developments, at the junction of climate science and economics. A large body of econometric studies analyzes the impacts of climate damages on the economy, with increasingly severe estimates of climate impacts (see Bilal and Känzig, 2026, which we draw upon to extend the model), though the present the rather belongs to the modelling part of the literature. Empirical results are nonetheless paramount to calibrate the models correctly, particularly on the damage functions, which we explain in more details in Section 3.4. Studies such as Weitzman (2010) and Dietz and Venmans (2019) have indeed been key to create a bridge between the empirical estimates of damages and their inclusion in macroeconomic models. On the modelling side, the literature has really taken off with the development of integrated assessment models (IAM), popularized by Nobel prize-winning William Nordhaus and his DICE model (Nordhaus, 1992), the latest version being that of Barrage and Nordhaus (2024), which has become the standard both in the literature and in policy work. IAMs are notably used by the Intergovernmental Panel on Climate Change (IPCC), the ultimate reference on climate policy, to make its projections (IPCC, 2023). Some macroeconomic frameworks have since tried to integrate the effects of climate change by including a "climate block" based on those IAMs, such as E-DSGE (which stands for "Environmental Dynamic Stochastic General Equilibrium") models, like Heutel (2012) and Annicchiarico and Di Dio (2015). Those models are becoming increasingly influential in policy circles, in central banks, thanks to the Network on Greening the Financial System (NGFS), but also in other institutions such as the IMF or the World Bank. The Seghini and Dees (2025) model we replicate here largely draws upon this literature.

The other established literature is that on public debt. Studies on the subject have mainly soared as a result of extreme situations that required a mix of empirical and theoretical approaches to inform policymakers. The knowledge on public debt has thus first been the knowledge on public debt crises. It has primarily been developed to analyze the case of developing countries, stemming from the numerous crises they had faced since the 1980s (Cohen and Sachs, 1986; Krugman, 1988; Calvo, 1988), and the linkages with other types of crises, such as liquidity and currency crises. But the theoretical developments that enabled institutions such as the IMF, the World Bank, and most central banks to set a common framework to analyse the issue, "Debt Sustainability Analysis" or "DSA", largely drew on the debate sparked by the European sovereign debt crisis of the 2010s. The context of analysis thus switched to developed countries, such as Portugal, Italy, Greece and Spain (a group of countries otherwise known under some rather unflattering acronym). France also suffered, especially compared to Germany, from this period, and the divergence between the two countries can largely be seen in the data (see Figure 2 in introduction).

Building on all these crises, two of the most renowned authors on the matter, Carmen Reinhart and Kenneth Rogoff analyzed historical data on the negative the impact of public debt on the economy, notably GDP growth (Reinhart and Rogoff, 2010). They came to the conclusion that public debt does not just reduce governments' margins of action, but that it provokes actual and potentially irreversible macroeconomic difficulties. What would thus be the optimal level of public debt ? when is public debt too much ? Reinhart & Rogoff find that beyond a 90% debt to GDP ratio (for developed countries, and 60% for emerging markets) growth starts to decline. This symbolic threshold was largely used by European policy makers as a justification to push austerity policies at a time when the economy was declining. If one intrinsically believes that exceeding a 90% threshold leads to the apocalypse, avoiding it at all costs may seem the most rational thing to do. This is what has been done and a few years later we realized that the exact inverse should have been adopted (Blanchard and Leigh, 2014). The patient may hardly recover from this "bleeding cure", in the words of Paul Krugman (Krugman, 2011). This idea of an exogenous debt threshold was also largely put into question as it was discovered that Reinhart and Rogoff (2010) proved to contain some miscalculations and that the threshold should have been higher, even though the authors, while admitting to it, stated that it did not invalidate their argument (Reinhart and Rogoff, 2013).

This debate led economists to study even more the question of debt sustainability. It has since then been acknowledged that is relies less on a debt to GDP threshold than on the key economic dynamics involved, namely the expected trajectory of the fiscal balance and the difference between government borrowing rates and the rate of growth of output (see equation (1)). This is why economists as Olivier Blanchard state

that a highly indebted country, such as France, can actually contract more debt as long as this interest-rate-growth gap is negative (Blanchard, 2019), which has been the case over most of the past decade (see bottom-left panel in Figure 2). The principle of debt sustainability analysis comes from a simple accounting identity, namely that current debt should be equal to the infinite sum of discounted future surpluses the sovereign is expected to generate. Obviously, one does not know exactly if or when a country will go out of deficit, and this also heavily relies on preferences for the present. This identity, being more about "expected" values, thus relies heavily on investor sentiment. A foundational work, which is the backbone of DSA frameworks (particularly the IMF's), is that of Ghosh et al. (2013), who, in a mix of theory and empirical analysis, account for these dynamics to provide for a set of countries a "debt limit" after which the government may enter a vicious circle leading it to default on its debt. Since then, DSA and the concept of "fiscal limit" has been a reference in the literature, influencing works such as Collard et al. (2015) that influenced the modelling of debt included in Seghini and Dees (2025). Public debt sustainability has also been largely integrated into standard macroeconomic models like DSGEs as in Corsetti et al. (2013), also one of the building blocks Seghini and Dees (2025).

A major challenge consists in combining those two established frameworks : climate change and debt sustainability. The literature is still very emergent on the topic. Most papers focus on sovereign credit worthiness, resorting to the empirical estimation of the "green premium" on government bonds (Battiston and Monasterolo, 2020; Klusak et al., 2023). Others, such as Zenios (2022), try to integrate the impact of climate change in the standard DSA framework. Some work can be found on the fiscal costs of climate change in Barrage (2020a) and Barrage (2020b). Central banks and international institutions have been starting to gain interest in incorporating the debt-climate nexus in their projections (Garcia-Macia et al., 2024). However, few peer-reviewed articles can be found on the matter, which suggest that there remains a sizable research gap to be filled. The only model to our knowledge to comprehensively combine the framework of E-DSGEs with that of public debt sustainability is Seghini and Dees (2025). This motivated our choice to replicate it as a starting point for our analysis.

3 Model

3.1 Hypotheses and logic of the model

We are taking the model of Seghini and Dees (2025), that we managed to replicate, and then complement with some extensions. It is actually not a DSGE (dynamic stochastic general equilibrium) model but rather a DGE (dynamic general equilibrium) since no stochastic aspect is involved. It belongs to the family of deterministic single-agent real business cycle (RBC) models. The macroeconomic variables involved are thus expressed in real terms (no monetary policy rule or Philipps Curve) and the agents are defined as homogeneous (one representative firm, one representative consumer) and assumed to behave in "perfect foresight", in the sense that they correctly anticipate the economic dynamics at play when making their choices, including the "permanent shock" represented by the green transition. The policy scenarios, used to evaluate transition impacts, are specified as "parameter combinations" rather than shocks. The model includes the physical damages of climate change on the economy via the "damage function" as well as a "fiscal bloc" in which government debt is considered as "risky".

The model poses a constraint on both CO2 emissions and public debt. It is assumed that the economy will transition no matter what to a "green state" in which emissions are zero and that public debt has been stabilized at its initial level. So the model does not answer to whether or not we will succeed in fighting climate change, but rather to whether, assuming we do, what is the resulting impact on the economy. The results aim to show what does the "green state" look like, what are the dynamics that lead to it, and how do they vary across the different possible ways - policy scenarios - of making the transition. Those scenarios are outlined and analyzed in Section 4.

3.2 Equations

The equations can be placed in four different blocks : Environment, Firms, Households and Government. To ensure smoother reading, we inform the reader here that the interest rate variables displayed in the model equations (R_t, R_t^d, R_t^k, that we define later in this section) are defined so as to have a value is superior to one, the equivalent of the $1 + r_t$ with $r_t \geq 0$ that may be found in other contexts.

3.2.1 Environment

As mentioned above, the model assumes that the transition is carried out in full : the sum of all national emissions E_t from the start date onwards must stay within the carbon budget $\bar{E}_0^n$ compatible with the corresponding warming target for our country,

as described in (E1). The budget is exhausted around 2060 in the 1.5°C scenario and around 2150 in the 3°C scenario, dates that follow from the mitigation rule (E4) rather than being imposed.

$$\sum_{t=0}^{+\infty} E_t \;\leq\; \bar{E}_0^n. \tag{E1}$$

We assume in (E3) that the country considered in the simulation is followed by the rest of the world in making the transition, so the global cumulative emissions H_t^G move linearly with national cumulative emissions H_t - defined in (E2) - with a slope equal to the ratio between the world carbon budget $\bar{E}_0^G$ and the national one $\bar{E}_0^n$.

$$H_t = H_{t-1} + E_t \quad \Longleftrightarrow \quad H_t = \sum_{j=0}^{t} E_j \tag{E2}$$

$$H_t^G = H_{-1}^G + \frac{\bar{E}_0^G}{\bar{E}_0^n} H_t. \tag{E3}$$

Furthermore, in (E4), the total mitigation effort, namely the sum of the private effort m_t and the public effort m_t^g, follows the part of the carbon budget that has already been spent, governed by a curvature parameter p. We also pose $\theta \equiv m_t^g / (m_t + m_t^g)$.

$$m_t + m_t^g \;=\; \left(\frac{H_{t-1}}{\bar{E}_0^n}\right)^p, \qquad p > 0 \tag{E4}$$

3.2.2 Firms

The production function (F1) of the representative firm i is a standard Cobb-Douglas with capital $K_t(i)$ whose output share is denoted by κ and labor $L_t(i)$ is scaled by an efficiency factor Z_t that represents technological progress, and grows at rate z, so that $Z_t = Z_0 e^{zt}$ and $Z_0 = 1$.

$$Y_t(i) = D_f(H_t^G)\, K_t(i)^\kappa \, [Z_t L_t(i)]^{1-\kappa}, \qquad \kappa \in (0,1). \tag{F1}$$

Total factor productivity $D_f(H_t^G)$ in each period is "damaged" as a result of the many consequences of climate change, all captured in the so call "damage function". This damaged productivity, on which we will come back extensively in Section 3.4, is expressed as a function of the stock of accumulated CO2 emissions H_t^G.

In (F2), the flow of emissions E_t is a byproduct of output $Y_t(i)$ in a proportion ξ_t, diminishing on its own at rate ω, so that $\xi_t = \xi_0 e^{-\omega t}$. It represents the decoupling of output from emissions that is projected to increase with technological innovation.

$$E_t(i) = \xi_t \, [1 - m_t(i) - m_t^g] \, Y_t(i), \tag{F2}$$

The emissions are mitigated by firms and government at the respective costs $A_t(i)$ and A_t^g respectively defined in (F3) and (F4). The cost structure of public abatement differs from that of the private sector in that it is less "convex", which means that at the beginning of the transition, private efforts are comparatively cheaper a the margin, whereas after a certain point, as the cost grows more rapidly, it eventually exceeds public cost (that remain stable at the margin).

$$A_t(i) = \theta_1 \left[m_t(i)\right]^{\theta_2} Y_t(i), \qquad \theta_1 > 0,\ \theta_2 > 1. \tag{F3}$$

$$A_t^g = \theta_3 \left(m_t^g\right)^{\theta_4} Y_t, \qquad \theta_3 > 0,\ 0 < \theta_4 < \theta_2. \tag{F4}$$

The profit of firms (F5), beyond including labor costs (W_t is the wage paid to labor) and rent (R_t^k is the rate at which private capital, owned by households, is rented to firms), has also to take into account the abatement effort $A_t(i)$ incurred by firms, that may be subsidized by the government at rate s_t^A as well as the carbon tax τ_t^E paid on each unit of CO2 emitted by the firm.

$$\Pi_t(i) \ = \ Y_t(i) \ - \ W_t L_t(i) \ - \ R_t^k K_t(i) \ - \ (1 - s_t^A) A_t(i) \ - \ \tau_t^E E_t(i), \tag{F5}$$

$$\Theta_t(i) \ \equiv \ 1 - (1 - s_t^A)\theta_1 m_t(i)^{\theta_2} - \tau_t^E \xi_t [1 - m_t(i) - m_t^g] \tag{F6}$$

In (F6) we define $\Theta_t(i)$ as the "wedge" of transition policy with respect to firm output, which can also be interpreted as the Lagrange multiplier associated to (F1), as in our real model the marginal cost of the firm is equal to 1 since perfect competition is assumed.

The first order conditions with respect to L_t, K_t and m_t are respectively outlined in (F7), (F8) and (F9). As all firms are homogeneous, we can drop the i in parentheses.

$$W_t = (1 - \kappa)\Theta_t \frac{Y_t}{L_t} \tag{F7}$$

$$R_t^k = \kappa\, \Theta_t \frac{Y_t}{K_t} \tag{F8}$$

$$\tau_t^E = \frac{(1 - s_t^A)\theta_1 \theta_2}{\xi_t} m_t^{\theta_2 - 1} \tag{F9}$$

3.2.3 Households

The optimization problem of the household is outlined in (H1), where utility is defined as logarithmic in consumption C_t and isoelastic with respect to labor, where σ_l stands for the inverse Frisch elasticity of labor supply, μ_L the disutility of labor and β the

household's discount factor.

$$\max_{\{C_t,L_t,B_t,I_t,K_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\ln C_t - \mu_L \frac{L_t^{1+\sigma_l}}{1+\sigma_l} \right], \qquad \beta \in (0,1),\ \mu_L > 0,\ \sigma_l > 0. \tag{H1}$$

The household's intertemporal utility defined in (H1) is optimized under two constraints.

The first is the budget constraint (H2), where I_t is the household's investment, B_t its government debt holdings that pay a R_t^d interest rate and Π_t the dividends it gets from firms (equal to zero in equilibrium) and T_t a lump sum tax (or transfer). The particularity in this model is that government debt is risky, and thus may default. The haircut to debt is denoted by ϑ_t. It is zero except when the government defaults in which case it is equal to $\bar{\vartheta}$. Over this haircut, a fraction f will be definitively lost, while the rest will be paid back to households in the form of a one time lump sump transfer $V_t = (1-f)\vartheta_t R_{t-1}^d B_{t-1}$. Consequently, (H2) can be rewritten as in (H3), where the terms are rearranged and we can see the "consolidated haircut" $f\vartheta_t$.

$$C_t + I_t + B_t \;=\; (1-\vartheta_t)R_{t-1}^d B_{t-1} \;+\; W_t L_t \;+\; \Pi_t \;+\; R_t^k K_t \;-\; T_t \;+\; V_t, \tag{H2}$$

$$C_t + I_t + B_t \;=\; (1-f\vartheta_t)R_{t-1}^d B_{t-1} + W_t L_t + \Pi_t + R_t^k K_t - T_t \tag{H3}$$

The second constraint is the capital accumulation constraint (H4), where δ stands for depreciation and S is a quadratic adjustment cost function defined in (H5), with ι a parameters guiding the scale of the adjustment costs.

$$K_{t+1} \;=\; (1-\delta)K_t \;+\; I_t \left[1 - S\left(\frac{I_t}{I_{t-1}} \right) \right], \tag{H4}$$

$$S(x) = \frac{\iota}{2}(x - e^z)^2, \qquad S'(x) = \iota(x - e^z), \qquad S''(x) = \iota > 0. \tag{H5}$$

Defining λ_t as the lagrangian multiplier associated to the budget constraint (H3) and λ_t^k as that associated to the capital accumulation constraint (H4), we can write the lagrangian function as in (H6).

$$\begin{aligned} \mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Bigg\{ & \ln C_t - \mu_L \frac{L_t^{1+\sigma_l}}{1+\sigma_l} \\ & + \lambda_t \Big[(1-f\vartheta_t)R_{t-1}^d B_{t-1} + W_t L_t + \Pi_t + R_t^k K_t - T_t - C_t - I_t - B_t \Big] \\ & + \lambda_t^k \Big[(1-\delta)K_t + I_t\big(1 - S(I_t/I_{t-1})\big) - K_{t+1} \Big] \Bigg\}. \end{aligned} \tag{H6}$$

The first order conditions with respect to C_t, L_t, B_t, K_{t+1} and I_t are respectively outlined in (H7), (H8), (H9), (H10) et (H11).

$$\lambda_t = \frac{1}{C_t} \tag{H7}$$

$$\mu_L L_t^{\sigma_l} = \lambda_t W_t \tag{H8}$$

$$\lambda_t = \beta R_t^d \, \mathbb{E}_t[\lambda_{t+1}(1 - f\vartheta_{t+1})] \tag{H9}$$

$$\lambda_t^k = \beta \mathbb{E}_t\Big[\lambda_{t+1} R_{t+1}^k + \lambda_{t+1}^k (1-\delta)\Big] \tag{H10}$$

$$\begin{aligned} \lambda_t \; = \lambda_t^k \left[1 - S\Big(\tfrac{I_t}{I_{t-1}}\Big) - S'\Big(\tfrac{I_t}{I_{t-1}}\Big) \frac{I_t}{I_{t-1}}\right] \\ + \beta \, \mathbb{E}_t\left[\lambda_{t+1}^k \, S'\Big(\tfrac{I_{t+1}}{I_t}\Big)\Big(\tfrac{I_{t+1}}{I_t}\Big)^2\right] \end{aligned} \tag{H11}$$

As a result, in (H12) we define q_t^k, Tobin's Q, as the ratio of the capital accumulation constraint multiplier (value of installed capital) and the budget constraint multiplier (marginal utility).

$$q_t^k \equiv \frac{\lambda_t^k}{\lambda_t} \tag{H12}$$

To find the risk-free rate, we take in (H14) the inverse of the stochastic discount factor, namely the present value of tomorrow's expected marginal utility in terms of today's marginal utility. Adding (H10) and (H12) we obtain (H13).

$$R_t = \frac{1}{q_t^k \, \mathbb{E}_t[\lambda_{t+1}]} \Big\{ \mathbb{E}_t[\lambda_{t+1} R_{t+1}^k] + (1-\delta) \, \mathbb{E}_t\Big[\lambda_{t+1} q_{t+1}^k \left(1 - \omega_q \, f\bar{\vartheta} \, \mathrm{PD}_{t+1}\right)\Big] \Big\} \tag{H13}$$

$$R_t \equiv \frac{\lambda_t}{\beta \mathbb{E}_t[\lambda_{t+1}]} \tag{H14}$$

Finally, we plug (H9) into (H14) and assume as an approximation the fact that marginal utility should be independent from the haircut and government interest rates to obtain in (H15) the "spread" or "wedge" between sovereign rate and the risk-free rate interest rate.

$$R_t \approx R_t^d \, (1 - f\bar{\vartheta} \cdot \mathrm{PD}_{t+1}), \tag{H15}$$

3.2.4 Government

in (G1), (G2), (G3) and (G4) we define the key equations underlying the dynamics of public debt accumulation, with B_t standing for the level of public debt in t at face value and D_t the level of public debt including interest due. S_t accounts for the primary surplus, which is the fiscal fiscal balance to which we add back the interest expense

(so that in the case of a deficit, it is smaller in size).

$$B_t = R^d_{t-1} B_{t-1} - S_t \tag{G1}$$

The structure of the primary surplus is detailed in (G2), whereby we subtract "normal" government expenses G (for which we fix the share of output in this version of the model) and environmental expenditures to the revenues collected from the lump sum tax T_t and those collected from the carbon tax. Expressed as a share of output, G and T_t become respectively g and τ_t.

$$\begin{aligned} S_t \;=\; \Big[& \tau_t + (1-\omega_g)\,\tau^E_t \xi_t (1 - m_t - m^g_t) - g \\ & - s^A_t \theta_1 m_t^{\theta_2} - \theta_3 (m^g_t)^{\theta_4} \Big] Y_t \end{aligned} \tag{G2}$$

$$d_t \equiv \frac{D_{t+1}}{Y_t} = \frac{B_t \cdot R^d_t}{Y_t} = b_t R^d_t. \tag{G3}$$

The equation of debt dynamics, alternatively expressed in equation (1) at the end of the introduction is translated in terms of the model in (G4), with $g^Y_t = Y_t / Y_{t-1} - 1$.

$$b_t = \frac{R^d_{t-1} B_{t-1}}{Y_t} - s_t = \frac{R^d_{t-1}}{1 + g^Y_t}\, b_{t-1} - s_t, \tag{G4}$$

In (G5), we define default as the case when the public debt to GDP ratio exceeds its debt limit b^M - a value taken from the literature - plus a log normal shock, which creates a grey zone in which default may occur even if b_t is slightly below its limit or may not occur even though the limit has been slightly exceeded. The volatility of the shock σ^2_B governs the size of this grey zone.

$$\text{Default}_{t+1} \iff b_t > b^M e^{\epsilon_t}, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma^2_B). \tag{G5}$$

As a result, we define the probability of a sovereign default in (G6).

$$\text{PD}_{t+1} \;=\; \Phi\left(\frac{\ln b_t - \ln b^M}{\sigma_B} \right) \tag{G6}$$

Eventually, (G7) and (G8) define the fiscal adjustment rule, according to which non carbon fiscal revenues τ_t rise automatically as debt rises or gets away from its target value (which in the calibration is identical to the starting value), the scale of this reaction being governed by ϕ_b. The reaction is slowed down by an "inertia" term $\phi_{\tau,t}$, that depends on the terminal tax rate $\tau^\star$, scaled by a parameter $\phi_{\tau,\star}$ and is increased in the presence of climate damages. The size of this increase is proportional to the initial GDP loss incurred - $y_{0,\text{NC}}$ being its value with no damage and $y_{0,n}$ its value with the damage - and the distance of yesterday's debt to GDP ratio to its target level.

$$\tau_t \;=\; \phi_{\tau,t}\big[\tau_{t-1} + \phi_b (b_t - b_{t-1})\big] \;+\; (1 - \phi_{\tau,t})\big[\tau^\star + \phi_b (b_t - b^\star)\big] \tag{G7}$$

$$\begin{aligned}\phi_{\tau,t} \;=\; & \frac{1}{1+\phi_{\tau,\star}\tau^{\star}} \\ & +\; \mathbf{1}_{\{D_f \neq \bar{D}\}} \cdot \phi_{\tau,y}\left(y_{0,\mathrm{NC}} - y_{0,n}\right)\left(b_{t-1} - b^{\star}\right).\end{aligned} \tag{G8}$$

Two choices of our implementation depart from this written form. First, in all the simulations shown, the first term of (G8) is not derived from $\tau^{\star}$ but held fixed at 0.904 (Table 5). Second, the anchor of (G7) is not the terminal tax rate $\tau^{\star}$ from the first period on, but a target that moves from the initial tax rate τ_0 to $\tau^{\star}$ in step with total abatement $M_t = m_t + m_t^g$, namely $\tau_0 + (\tau^{\star} - \tau_0)\,(M_t - M_0)/(1 - M_0)$: with a fixed anchor, the scenarios in which the state carries most of the abatement have no solution.

Finally, the market has to clear, as shown in (M).

$$Y_t \;=\; C_t \;+\; I_t \;+\; G_t \;\;+\; A_t \;+\; A_t^g \tag{M}$$

3.3 Implementation

3.3.1 Detrending and steady state

All the model equations above have been defined in levels. However, the model, as in every DGE, reasons in deviations around a long term trend rather than in absolute values. Consequently, all its equations have to be "de-trended". This means that in the simulations, all the variables that contain a trend, such as Y_t, C_t, I_t, K_t, W_t, B_t, D_t, T_t, S_t, G_t, A_t, A_t^g, λ_t and λ_t^z, that grow in the long run at the rate of growth of exogenous technological progress z, ξ_t at minus the rate of growth of emissions-output decoupling ω and E_t, H_t and H_t^G at the difference between those two rate, $z - \omega$, have to be divided by the corresponding trend factor e^{zt}, $e^{-\omega t}$ or $e^{(z-\omega)t}$. It happens however that as $z = \omega$ in the calibration, which highly simplifies the calculations and the handling of the environment bloc, notably by making emission flows stationary.

Once the de-trending has been carried out, one needs to obtain the steady state. One might then ask what is actually the steady state of our model, since we have a "brown state" (no mitigation, maximum emissions) at the beginning of the simulation and a "green state" (full mitigation, no emissions) at the end. A common practice in DGEs and DSGEs is to have a single steady state, such that the economy, which is in equilibrium at the beginning of the simulation, receives a shock, whose dynamic effect on all variables of the model is actually what we seek to analyze, and eventually converges back to its initial equilibrium. This does not work in our model, in which the existence of two states necessarily different from one another (as implied by the green transition) breaks this logic. The economy cannot converge back to the "brown state". In fact, it happens that the "brown state" is not really a steady state, as it contains the emissions accumulation equation (E2), that is only stationary when $E_t = 0$, otherwise displaying a unit root. As a result, the only true steady-state in the model is the green

terminal state. The brown state is therefore only regarded as an "initial condition" set for all variables (this has an impact on the simulation architecture that we build in matlab/dynare).

3.3.2 Calibration

The calibration we used, outlined in Tables 1 to 5, draws from Seghini and Dees (2025) and Seghini et al. (2026) unless otherwise specified.

Table 1: Preferences and technology

Symbol	Value	Description
β	0.995	Discount factor
κ	0.33	Capital share in production
δ	0.025	Depreciation rate
σ_l	1	Inverse Frisch elasticity of labour supply
μ_L	19.841	Weight of the disutility of work (hours worked about one third of the time endowment)
ι	15	Investment adjustment cost
z	0.005	Trend growth rate of labour efficiency Z_t (2 % per year)
ω	0.005	Decline rate of carbon intensity, set equal to z
$\bar{D}$	1	Productivity level in the absence of damage, normalised so that $D_f = 1$ at the initial warming $\widehat{T}_0$ under every damage specification

Table 2: Abatement and emissions

Symbol	Value	Description
θ_1	0.023	Scale of private abatement cost, share of output
θ_2	1.8	Convexity of private abatement cost
θ_3	0.023	Scale of public abatement cost, share of output
θ_4	1.044 (asymmetric costs), 1.8 (symmetric costs)	Convexity of public abatement cost, $\theta_4/\theta_2 = 0.58$ or 1
ξ_0	0.17	Initial carbon intensity of unabated output, $GtCO_2$ per unit of output
p	0.5 (1.5 °C world), 3.0 (3 °C world)	Curvature of the mitigation rule

Table 3: Carbon budget and climate

Symbol	Value	Description
$\bar{E}_0^G$	240 (1.5 °C world), 3055 (3 °C world)	World carbon budget, $GtCO_2$
$\bar{E}_0^n$	1.9 (1.5 °C world), 24.2 (3 °C world)	National carbon budget, $GtCO_2$, the 3 °C value keeping the national share of 1.9/240
H_{-1}^G	2650	World cumulative CO_2 emissions at the start date, $GtCO_2$ (Seghini et al., 2026)
λ_{TCRE}	0.00053	Warming per $GtCO_2$, °C (Seghini et al., 2026)
$\widehat{T}_0$	1.4	Warming above pre-industrial at the start date, °C, equal to $\lambda_{\text{TCRE}} H_{-1}^G$

Table 4: Damage specifications (as a function of warming $\widehat{T}$ above pre-industrial, see Section 3.4)

Symbol	Value	Description
D_1	$1 - 0.003467\,\widehat{T}^2$	DICE-2023 (Barrage and Nordhaus, 2024)
D_2	$1 - 0.0335\,\widehat{T} - 0.0009\,\widehat{T}^2$	Kalkuhl and Wenz (2020)
D_3	$\exp(-0.01\,\widehat{T}^2)$	Seghini and Dees (2025), $\gamma_3 = 0.02$, $\gamma_4 = \lambda_{\text{TCRE}}$
D_4	$\exp(-0.04\,\widehat{T}^2)$	Seghini et al. (2026),$\gamma_3 = 0.08$
D_5	two-kernel dynamic specification	Bilal and Känzig (2026), see Section 3.4.2

Table 5: Fiscal block

Symbol	Value	Description
g	0.0575	Public consumption to output
$b^\star$	1.10	Debt to output target, equal to the initial debt ratio
b^M	1.35	Fiscal limit, maximum sustainable debt to output
σ_B	0.08	Dispersion of the fiscal limit
$\bar{\vartheta}$	0.70	Haircut in case of default
f	0.30	Share of the default loss borne by the financial sector
ϕ_b	0.2	Response of the tax rate to debt (G7)
$\phi_{\tau,t}$	0.904	Inertia of the fiscal rule, held fixed in all simulations shown (see Section 3.3.2)
$\phi_{\tau,y}$	0.9	Interaction between climate damage and debt in the rule inertia (G8)
ω_q	0 (baseline), 0.5 (spillover simulations)	Strength of the sovereign risk spillover onto capital (H13)

3.4 Adding a new damage function

3.4.1 Damage functions in the Seghini & Dees model

A damage function if a function that aims at capturing the relationship between global warming and all the material damages its entails for the economy. Usually, the output of this function is used as a multiplicative wedge to total factor productivity, although other forms exist (Weitzman, 2010). We stick here to the traditional form.

Moreover, although global warming is measured in the majority of cases with excess average global temperatures, our model happens to use cumulative CO2 emissions. Fortunately, climate scientists have designed a way to bridge those two measures : the transient climate response to emissions (TCRE). Indeed, Matthews et al. (2009) show that, over the long haul, the relationship between excess temperature and cumulated emissions becomes approximately linear. This result has been used notably by Dietz and Venmans (2019), who make the TCRE directly usable in damage functions in order to facilitate their use into climate-economy models. This tool is highly useful as it enables the modeler to avoid modelling the carbon cycle, even if it remains an approximation. For the coefficient λ_{TCRE} in $\widehat{T}_t \approx \lambda_{\text{TCRE}} . H_t^G$, where T_t is the excess temperature, they give a range between 0.000240 and 0.000749, for a central value of 0.000480. Seghini and Dees (2025) were using the upper value 0.000749. We use 0.00053 as in Seghini et al. (2026).

The original Seghini and Dees (2025) model uses two damage function specifications : quadratic (DQ) and exponential (DE). The quadratic damage function used in Seghini and Dees (2025) is the the one used in Annicchiarico and Di Dio (2015) and Heutel (2012).

$$D_{\text{quad}}(H_t^G) = \bar{D}\left(1 - \gamma_0 - \gamma_1 H_t^G - \gamma_2 (H_t^G)^2\right) \tag{DQ}$$

This quadratic specification is nonetheless most often expressed in the literature in terms of temperature rather than cumulative emissions (Barrage and Nordhaus, 2024; Kalkuhl and Wenz, 2020), which we retain in this study, to give a better benchmark of the "standard" or "consensus" found in the literature on damage functions. Our calibration thus implies a conversion that builds on the TCRE coefficient, as shown in (DQ*).

$$D_{\text{quad}}(\widehat{T}_t) = \bar{D}\left(1 - \gamma_0 - \frac{\gamma_1}{\lambda_{\text{TCRE}}}\widehat{T}_t - \frac{\gamma_2}{\lambda_{\text{TCRE}}^2}\widehat{T}_t^{\,2}\right) \tag{DQ*}$$

This standard, however, as Seghini & Dees recognize, may understate the scale of damages, which is why a second form, the exponential shape drawn from Dietz and

Venmans (2019), shown in (DE), is tested.

$$D_{\exp}(H_t^G) = \bar{D}\ \exp\{-\tfrac{\gamma_3}{2}(\gamma_4 H_t^G)^2\} \tag{DE}$$

We use the coefficients from Seghini and Dees (2025), but also those from Seghini et al. (2026), that were rescaled to match the damage found in the most recent literature, namely that of Bilal and Känzig (2026).

3.4.2 The Bilal & Känzig damage function

Bilal and Känzig (2026), estimating the impact of global temperatures on the local economy - instead of local temperature, which was what all the literature had been doing so far -, concluded that the impact of global warming on GDP was 10 times larger than the literature consensus. Our model's damage functions also happen to use global temperature to evaluate the physical damages caused by climate change on the national economy.

Although the DV function already incorporates more damage than the literature standard, it might still not match the x10 times of Bilal and Känzig (2026). Indeed, as mentioned above, in Seghini et al. (2026), it is the one function that has been retained, and the magnitude of the coefficients has precisely been rescaled (by a factor 4) to (approximately) match the damages seen in Bilal and Känzig (2026). However, the shape of Bilal & Känzig's damage function is quite peculiar, as it does not really follow the standard shapes that we are used to. We thus wondered about the impact produced by this shape rather than by a simple recalibration. This may be a way to come up with "more realistic" simulations of the impact of climate change on the economy.

$$D_{f,t} \;=\; \bar{D} \cdot \exp\left(\int_0^t \zeta_s\, \widehat{T}_{t-s}\, ds\right) \tag{3}$$

$$\zeta_s \;=\; A\big(e^{-Bs} - e^{-Cs}\big), \qquad C > B > 0 \tag{4}$$

It is indeed not expressed as a simple function of temperature (that can be converted to cumulative emissions thanks to the TCRE) or a simple exponential of the level of warming. Instead of the level of warming, we are making the assumption that warming has a dynamic effect (as opposed to the static effect of our previous functions), in the sense that a warming shock in t does not only have an effect on total factor productivity in t, but also on productivity in $t+1$, $t+2$ and so on. This is why current productivity is also related to past excess temperatures, obviously not to the same intensity as current temperature but to some degree still. The integral $\int_0^t \zeta_s\, \widehat{T}_{t-s}\, ds$ inside our exponent is thus nothing more than the continuous sum of the contemporaneous effect of $\widehat{T}_t$ and the past effects of $\widehat{T}_{t-s}$. Those effects are materialized by ζ_s, which is the

weight of each shock in period $t - s$, also called in mathematical terms the "kernel" of the convolution.

Now, the authors chose to impose a form on that kernel, a sum of two exponentials, which is why we may refer later to this specification as "two dimensional kernel" or "two-kernel". They imposed this form that relies on three parameters, A, B and C, calibrated from the results of their empirical estimation of the effect of global temperature shocks on output. We are thankful to Caterina Seghini for sharing with us the values of those parameters, that she managed to obtain from Diego Känzig.

How does it work ?

The form imposed by the authors on ζ_s amounts to an assumption on the behavior of this dynamic damage function, that could be of three types : transitory level effects, persistent level effects or growth effects. Transitory effects mean that the marginal time effect of a temperature shock ζ_s converges to zero over time. A transitory level effect means that this convergence happens very quickly, which would approximately amount to a static effect, described in our previous damage functions. A persistant level effect means that this convergence to zero happens sufficiently late in time for the "compounding effect" of the shock to be significant. Finally, a growth effect means that ζ_s converges to a strictly positive value, which means that every shock has a permanent effect over time, and thus does not only damage the level of GDP but its rate of growth. This last configuration is the most "damaging" for the economy. Indeed, level damages, as large as they can be when multiple temperature shocks occur in a row and compound themselves, can always theoretically be offset in the long run by the growth trajectory of the economy (the steady-state growth we evoked earlier, in the case of our model), whereas the growth damages have such a strong effect that only one single shock will mark the economy forever.

Although the authors don't formally exclude the presence of growth effects from their empirical analysis, they nonetheless settle for persistent level effects for the form of ζ_s, calling it a "conservative hypothesis". This choice is also obviously motivated by the need to obtain a steady-state for their structural model, which cannot exist in the presence of growth effects (which would more amount to a "full collapse state" than a "steady state", and even with that would not numerically converge). However, a debate is still ongoing in the literature to know whether climate change has growth effects or not, although the damage function used are still calibrated on local temperature (see Burke et al., 2015 and Dell et al., 2012), which makes this configuration seem all the more worrying in the case it proves to be closest to reality.

3.4.3 How we adapted it to fit the model

Simple recalibration

As we have just seen, accounting for the true dynamics of climate effects in our model involves a price : mathematical complexity. The convolution integral of Bilal and Känzig (2026) doesn't seem indeed to directly fit in our model, even of we are able to convert temperature shocks to cumulative emissions. One way to proceed and avoid this complexity if to simply take a damage function that already fits the model and re-calibrate its parameters to fit the desired impact. This is what the authors did in Seghini et al. (2026) by multiplying the γ_3 term of the exponential specification by a factor 4. Even though this is the safest way to proceed to emulate the impact, it does miss the dynamic shown by Bilal & Känzig in their paper, namely that effect of a temperature shock reaches a peak after two years (with a 4% productivity loss) after gradually phasing out after 10 years. We thus tried to import the two kernel form of Bilal & Känzig to the Seghini & Dees model to see how it compares to the authors' recalibrated version, see if we obtain more acute damages or not, and if yes how much more.

Two-kernel structure

First, the form imposed on ζ_s implies that we have

$$\log D_{f,t} = A \cdot (X_t^B - X_t^C) \tag{5}$$

with

$$X_t^B \equiv \int_0^\infty e^{-Bs} \cdot \widehat{T}_{t-s}\, ds, \qquad X_t^C \equiv \int_0^\infty e^{-Cs} \cdot \widehat{T}_{t-s}\, ds \tag{6}$$

X_t^B and X_t^C are both exponentially weighted averages of past warming. As $B < C$, X_t^B stands for the long-memory average (past shocks still count) and X_t^C for the short-memory average (past shocks fade quickly). Since $e^{-Bs} \geq e^{-Cs}$ for every $s \geq 0$, we always have $X_t^B \geq X_t^C$, and the damage is $A\,(X_t^B - X_t^C) \leq 0$ with $A < 0$. Intuitively, the two states evolve as follows after a warming shock: both start from zero, since the kernel is zero on impact ($\zeta_0 = 0$), and the gap $X_t^B - X_t^C$ then widens, reaches a maximum after $s^\star = \ln(C/B)/(C-B)$ years (about two years with the estimated parameters, in line with Bilal & Känzig's impulse responses) and narrows afterwards, as the short memory X_t^C fades quickly while the long memory X_t^B keeps a certain level and decreases only slowly over time.

Following the way climate scientists decompose the atmospheric carbon impulse response into a sum of exponentials (Strassmann and Joos, 2018), it can be shown that

these averages evolve according to the following behavior :

$$\dot{X}_t^B = \widehat{T}_t - B\, X_t^B, \qquad \dot{X}_t^C = \widehat{T}_t - C\, X_t^C \tag{7}$$

which, in a discretized version gives the following (with $\Delta t = 0.25$ if we choose a quarterly frequency, the parameters A, B and C having been calibrated annually) :

$$X_t^B = e^{-B\Delta t}\, X_{t-1}^B + \widehat{T}_t \cdot \frac{1 - e^{-B\Delta t}}{B} \tag{8}$$

$$X_t^C = e^{-C\Delta t}\, X_{t-1}^C + \widehat{T}_t \cdot \frac{1 - e^{-C\Delta t}}{C} \tag{9}$$

The whole mathematical derivation is consigned in Appendix A.

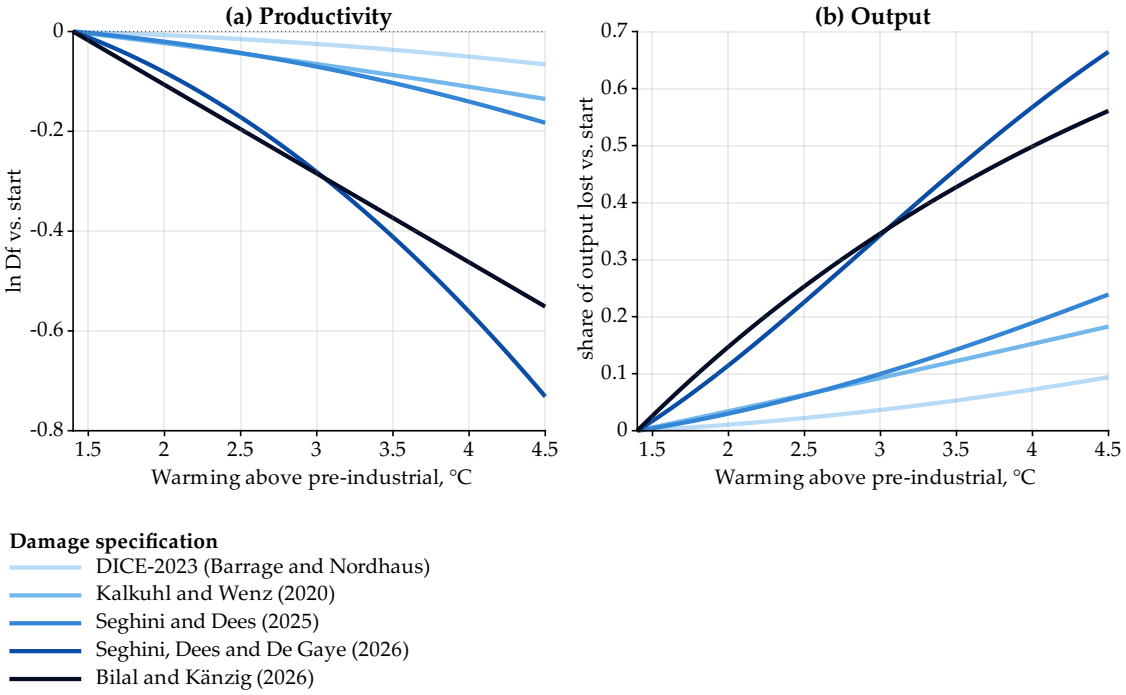


Figure 3: The five damage specifications as a function of warming

Let's now have a look at all those damage functions. We can see the shape of those functions with respect to temperature in Figure 3. The left-hand side panel thus shows value of TFP in logarithms while the right-hand side panel shows the share of output lost. Each damage spec on shown on the graphe has been normalized to its 2024 level (corresponding to a 1.4°C warming). The loss on production is obtained by taking the TFP damage in levels to the power $1/(1-\kappa)$, which accounts for the amplification of capital in the long run.

We thus take five different damage specifications, that will be used later in our analysis (see Section 4.1.2) : a first one (D_1), quadratic, from Barrage and Nordhaus (2024), that is the least severe and thus represents the lower bound on the spectrum of

damages in the current state of literature. A second one (D_2), quadratic as well, but more severe, resulting from the Kalkuhl and Wenz (2020) meta analysis on damage functions, from which we took the coefficients recommended by the authors to be plugged in integrated models. The third function (D_3) is simply that of Seghini and Dees (2025), taken from Dietz and Venmans (2019) with its original calibration and the fourth function (D_4) the recalibrated version found in Seghini et al. (2026). Finally, the fifth and last specification (D_5) is the "faithful" Bilal and Känzig (2026) function, which we approximated in this graph by the "long run" version which considers temperature $\hat{T}_t$ to be in steady state $\hat{T}$ in order to have a damage that can be expressed as an explicit function of $\hat{T}$, so that we have $X_t^B = \hat{T} \int_0^\infty e^{-Bs} = \hat{T}/B$ and in the same fashion $X_t^C = \hat{T}/C$, so that $\log D_f = A(1/B - 1/C)\,\hat{T} = -0.178\,\hat{T}$.

$$D_1(\hat{T}) = 1 - 0.003467\,\hat{T}^2 \tag{10}$$

$$D_2(\hat{T}) = 1 - 0.0335\,\hat{T} - 0.0009\,\hat{T}^2 \tag{11}$$

$$D_3(\hat{T}) = \exp\{\,-0.01\,\hat{T}^2\,\} \tag{12}$$

$$D_4(\hat{T}) = \exp\{\,-0.04\,\hat{T}^2\,\} \tag{13}$$

$$D_5(\hat{T}) = \exp\{\,-0.178\,\hat{T}\,\} \tag{14}$$

We notice that Kalkuhl & Wenz and Seghini & Dees are comparable, both definitely above Nordhaus & Barrage. Even though Kalkuhl & Wenz seems to be a tiny bit higher (although this is barely visible) until 2.5°C above pre-industrial levels, the curves cross and Seghini & Dees clearly takes the lead after 3°C. A similar phenomenon occurs with Seghini, Dees & De Gaye and Bilal & Känzig, which are by the way clearly the two most severe damage functions, largely above all the others. Bilal & Känzig remains above Seghini, Dees & De Gaye until 3°C, after which Seghini, Dees & De Gaye becomes the most severe damage function.

4 Results

4.1 Simulation results

4.1.1 The impact of transition policies

The benefit of a simulation approach is indeed to be able to simulate different trajectories against a chosen baseline, and just vary some selected parameters and see what happens. One can for instance try to see what is the effect of transition policies alone, which enables to say, when damages are added, what is the impact of those damages (one of the approaches suggested in the Pisani-Ferry and Mahfouz (2023) report). This is exactly what we do here, by considering first a framework without climate damages ("no climate", or "NC" scenario).

Table 6: Policy scenarios

Scenario	θ	s^A	ω_g	θ_4/θ_2
1. Carbon tax	0	0	0	0.58
2. Carbon tax, revenue recycled	0	0	0.25	0.58
3. Carbon tax and subsidies	0	0.9	0	0.58
4. Public investment, asymmetric costs	0.99	0	0	0.58
5. Public investment, symmetric costs	0.99	0	0	1

We consider 5 different scenarios (Table 6), which amount to different combinations of the public share of mitigation effort θ, allocation of carbon revenues to support government spending (ω_g) subvention to private abatement s^A and the ratio between the curvature of public abatement costs to that of private abatement costs (θ_4/θ_2). In scenario 1 ("Carbon Tax"), the full mitigation effort is borne by the private sector ($\theta = 0$), no carbon tax revenue is recycled ($\omega_g = 0$) so that it all goes to improving the government fiscal balance, no subventions are given to private abatement ($s^A = 0$) and the structure of public versus private abatement costs is assumed to be asymmetric, the public cost being less convex than the private one ($\theta_4 < \theta_2$) as evoked before (Section 3.2.2). In scenario 2 ("Carbon Tax, revenue recycled"), we allow some carbon tax revenue to be redirected to public spending g ($\omega_g = 0.25$), while in scenario 3 ("Carbon tax and subsidies") we chose instead to allow for massive subsidies to private abatement ($s^A = 0.9$). Scenario 4 ("Public investment, asymmetric") raises the share of the public mitigation effort to almost 100% ($\theta = 0.99$) instead of resorting to revenue recycling and subsidies and scenario 5 picks up the same configuration as scenario 4 but adding the assumption that the public and private abatement costs are identical ($\theta_4 = \theta_2$).

What we were able to replicate for the 1.5 °C world

We were thus able to qualitatively replicate the results obtained in Seghini and Dees (2025) under the 1.5 degree scenario (see Figure 4 and Figure 5). The only difference is that, as mentioned in Section 3.3.2, we evaluate the carbon budget only using CO2 emissions and not those of other gases (as was done in Seghini and Dees, 2025) to stick to the new calibration proposed by the authors in Seghini et al. (2026), so the temperature reached at the steady state is not "exactly" 1,5°C but 1,53°C - the considered scenario being exactly labelled as "50% probability of staying under 1,5°C". In the same way, the 3°C scenario evoked later corresponds to exactly 3.02°C reached.

In short, resorting only to a carbon tax to achieve the green transition (scenario 1) is safe fiscal-wise (debt, interest rates and default probability don't leave their "safe zone") and even leads to a drop in debt to GDP and default probability, which eventually rises back to go back to the target debt level (same as the one at the beginning - a calibration assumption). It nonetheless has a cost, as the high carbon tax rate paid by companies along the transition hampers investment, GDP and consumption. Recycling part of carbon tax revenues into public spending (scenario 2) barely changes those dynamics, the benefits and costs being just a little "less" pronounced. What radically changes the results, as one might have expected, is the switch from "full private financing of mitigation" to "(almost) full government financing". The logic becomes reversed, as fiscal variables quickly become under stress, as materialized by the peaks observed on debt to GDP, government interest rate and probability of default. This stress is most exacerbated under asymmetric abatement costs (scenario 4), whereas those peaks appear much more moderate under symmetric costs (scenario 5). This stresses out the role of the cost structure of abatement costs in choosing the "optimal" mix between public and private financing. Seghini and Dees (2025) indeed conclude, after having conducted a dynamic optimization of welfare, that it the optimal mix would consist of a high private share in the beginning of the transition progressively caught up by the public share once the private costs start to significantly increase. And the excess fiscal stress caused by this difference in cost structure is far from being compensated by a gain in GDP, investment and consumption, as the curves of scenarios 4 and 5 almost overlap (albeit a little less for consumption). Substituting subsidies for public abatement produced a kind of "middle ground" in which GDP, investment and consumption remain rather close to the "public mitigation" scenario dynamics, while drastically reducing the fiscal stress.

What we see in a 3°C degree world

As we can see in Figure 6, the impacts are not drastically different from the 1.5°C case. This is mainly due to the fact that we don't account yet for climate damages. Under the carbon tax scenario, consumption, output and investment experience a slightly stronger

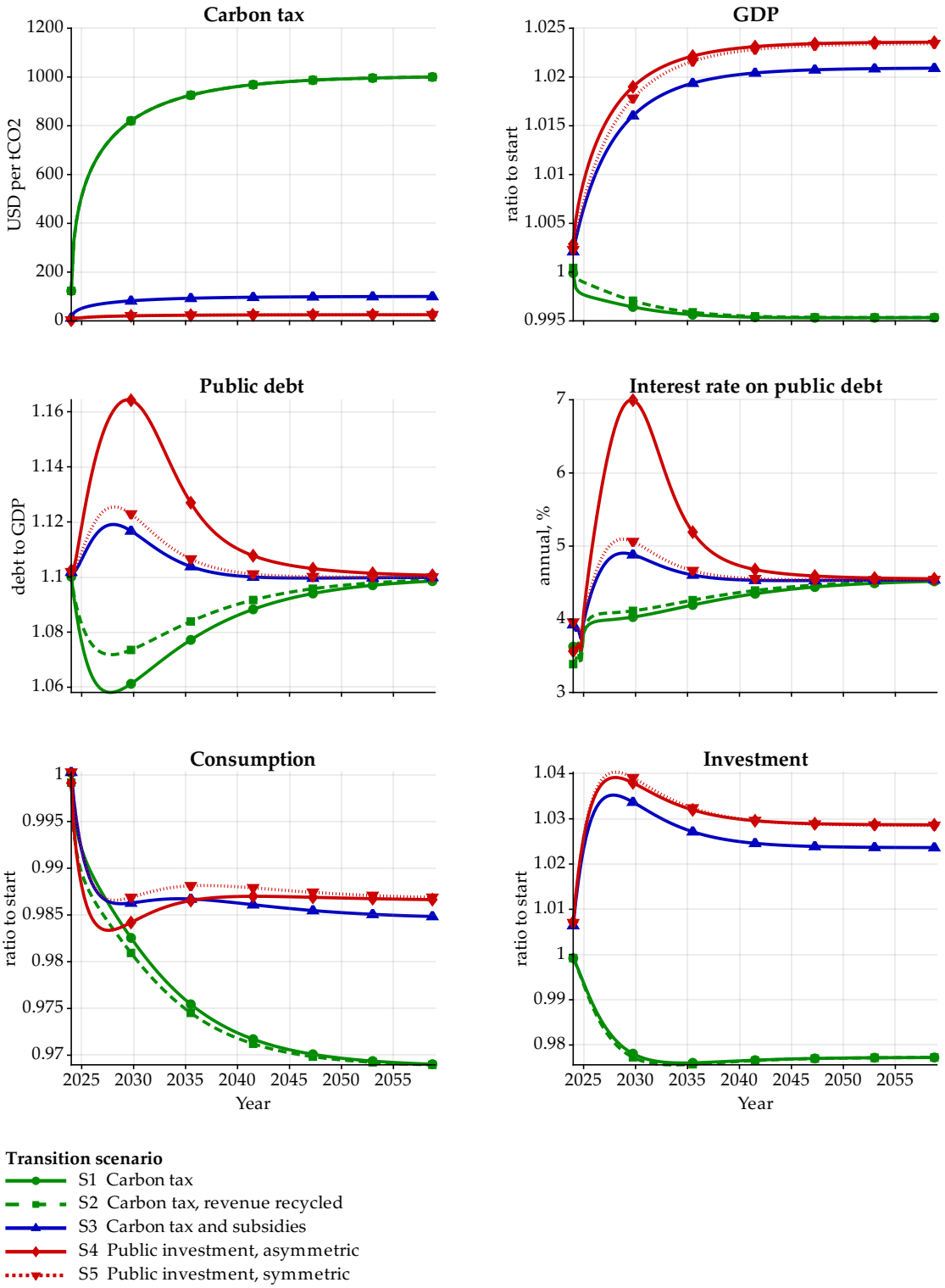


Figure 4: Macro-financial effects of the five policy scenarios, 1.5 °C world, no climate damage

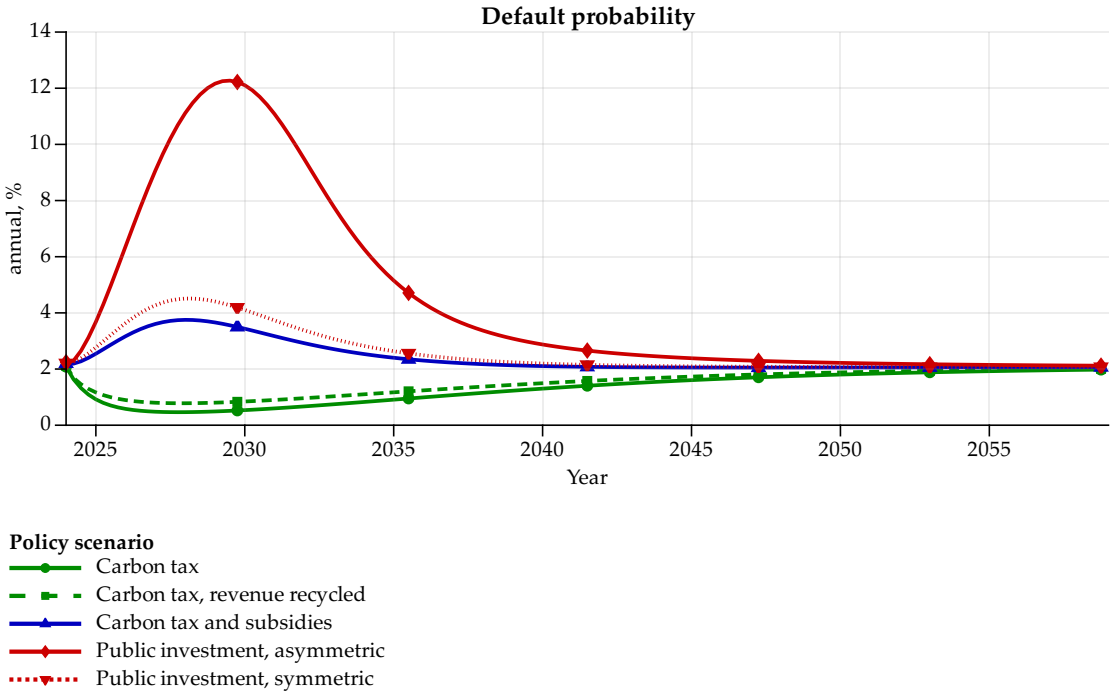


Figure 5: Annual default probability under the five policy scenarios, 1.5 °C world, no climate damage

albeit more progressive fall, while the dynamics of the "expansionary" scenarios are also more progressive, though less positive in the end for output and investment, and even in decline for consumption. The case of fiscal variables is very peculiar, since all the fiscal stress from the expansionary scenarios has now been "absorbed" by the larger carbon budget, and now all scenarios (except the S4 where the curve remains flat) show a trough in debt, which used to be only the case for the carbon tax scenarios under the 1.5°C budget.

4.1.2 The impact of climate damages

We illustrate the behavior of GDP, debt to GDP and default probability for our 5 scenarios, under 5 different damage specifications (Figure 7).

This time again, in the 1.5°C world, we find qualitatively the same patterns as in Seghini and Dees (2025). We can see that the damages largely affect GDP (this is also true of consumption and investment, although not represented in this graph) in all policy scenarios. For the fiscal variables, they are almost unchanged (the carbon tax scenario curves don't move, even under the strongest damage specifications), except in the public mitigation scenario with asymmetric costs, where the most severe specifications add to the fiscal stress, and the Bilal & Känzig one sends it through the roof: the peak default probability rises from 12.3% without damage to 13.8% under Seghini, Dees & De Gaye and 21.2% under Bilal & Känzig. The trajectory of GDP

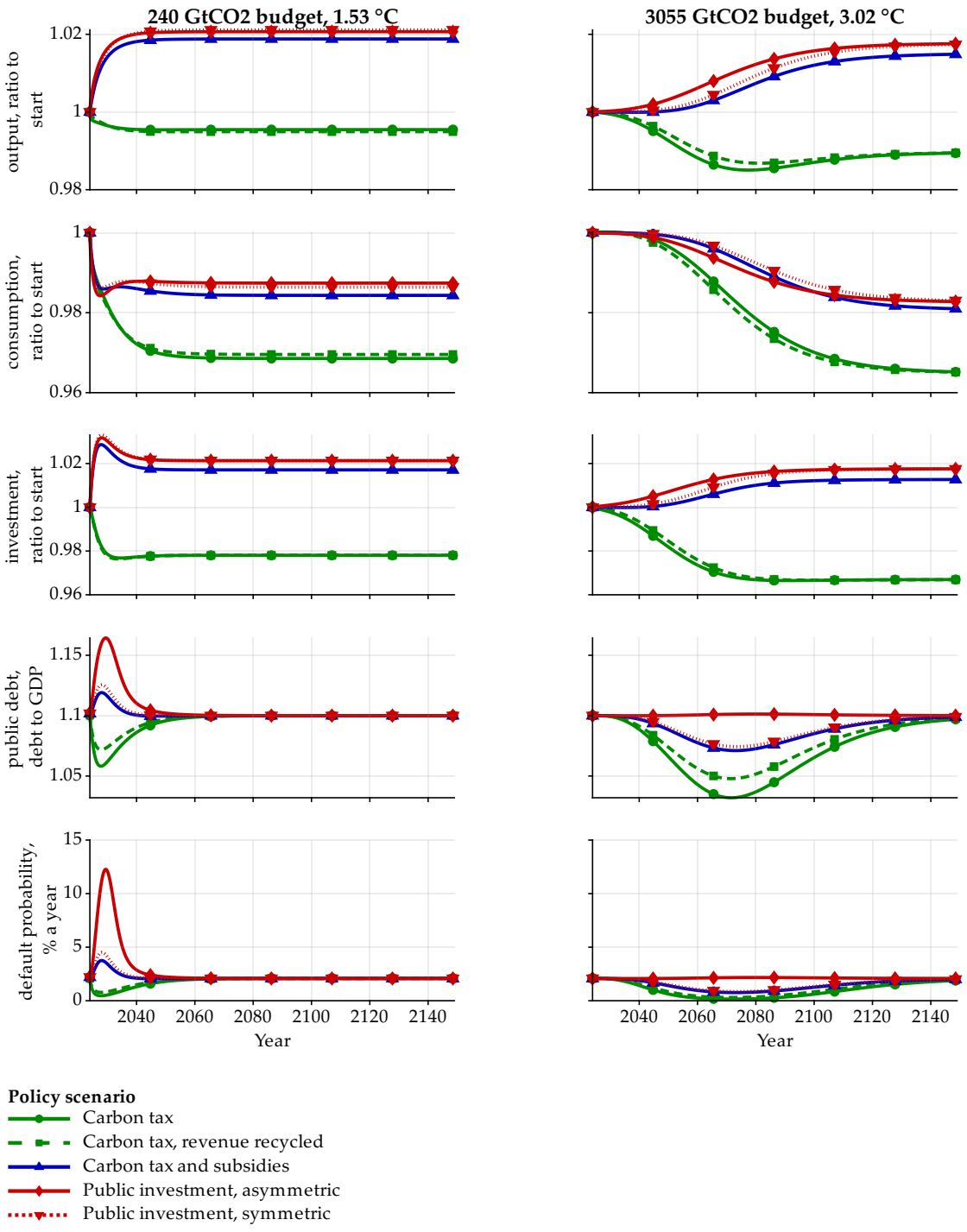


Figure 6: The five policy scenarios in the 1.5 °C and 3 °C worlds, no climate damage

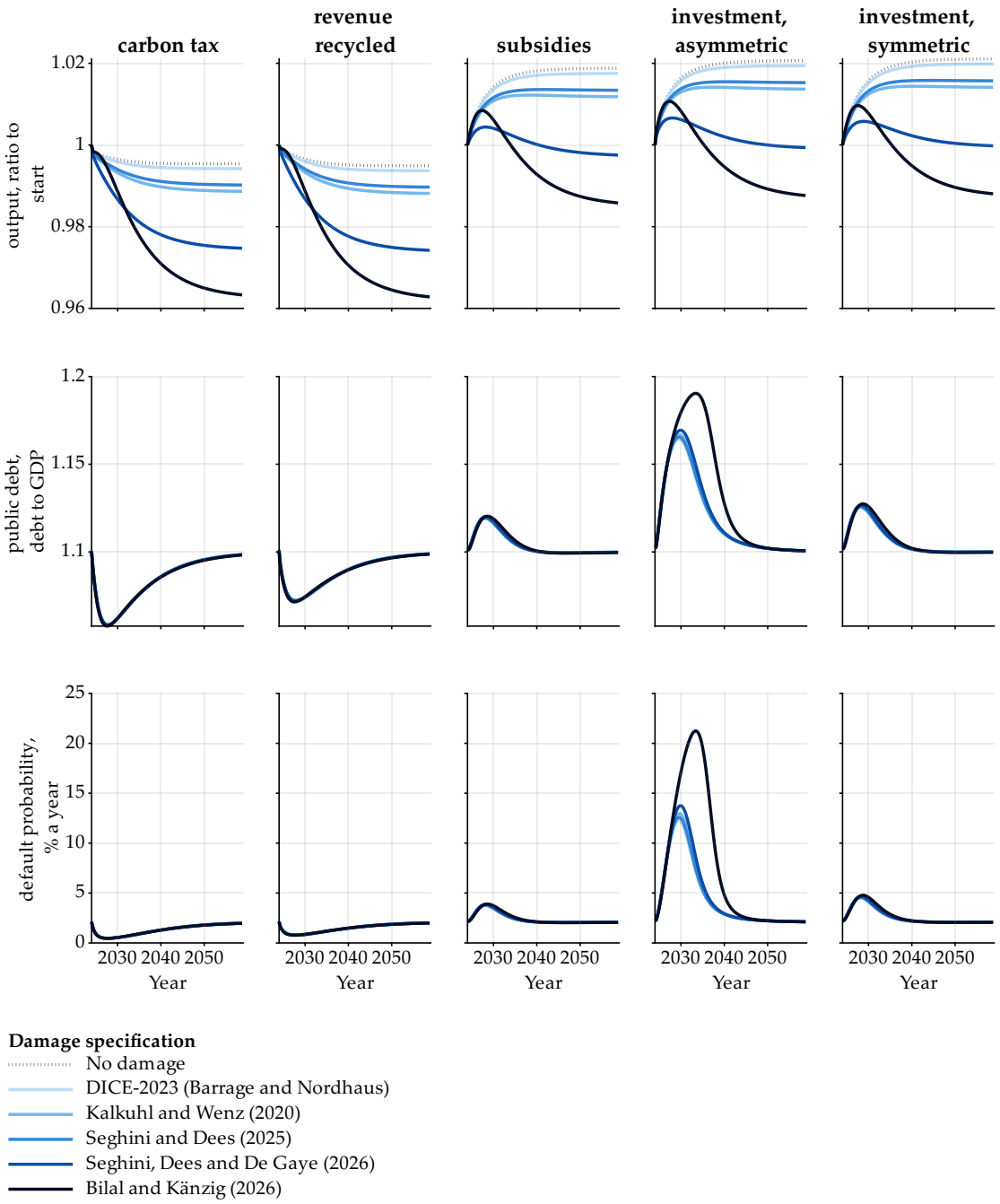


Figure 7: GDP, public debt and default probability under the five damage specifications, 1.5 °C world

shows the range of damages. The damage from DICE-2023 is very close to the no damage scenario, Kalkuhl & Wenz is slightly higher than that of Seghini & Dees, but Seghini, Dees & De Gaye is already much higher. At the terminal state, its additional impact with respect to Kalkuhl & Wenz is twice the difference between Kalkuhl & Wenz and the no climate scenario. Bilal & Känzig is on an even higher level: in the carbon tax scenario, output ends the window 3.2% below the no damage path, against 2.1% under Seghini, Dees & De Gaye. Those orders of magnitude can be found in each of the five policy scenarios.

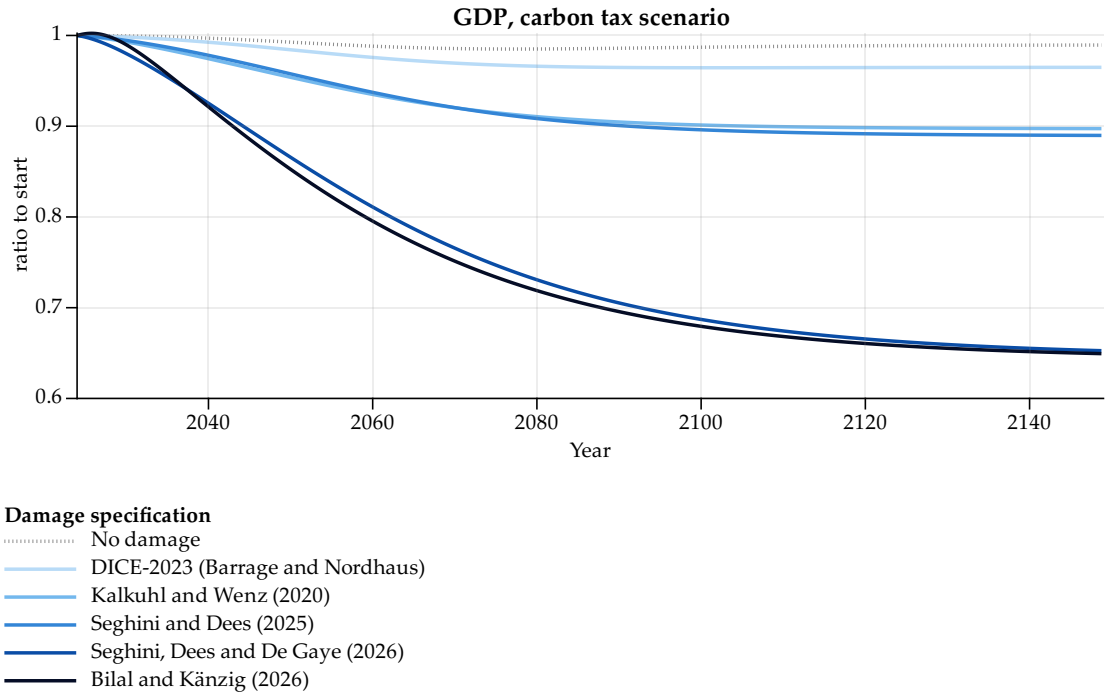


Figure 8: GDP under the five damage specifications, carbon tax scenario, 3 °C world

In a 3°C world (Figure 8), we notice that output is much more severely damaged than under 1,5 degrees. In the carbon tax scenario, the terminal loss in GDP is around 3% in the 1,5°C world for the most severe damage specification whereas it produces a loss of about 34% in the 3°C world, a 10-fold impact. The gap between the middle-ground "literature consensus" specifications (Kalkuhl & Wenz and Seghini & Dees, with losses of 9 to 10%) and the Bilal & Känzig specification is more than 3-fold, while the recalibrated specification of Seghini, Dees & De Gaye ends up almost exactly as severe as Bilal & Känzig, as the crossing of the two curves at 3°C in Figure 3 suggests. Consumption and investment, not shown here, follow the same dynamic as GDP, and the four other policy scenarios display the same pattern, with almost identical losses relative to their own no damage path.

We do not show the fiscal variables under climate damages in the 3 °C world, as our results there lack economic sense. Indeed, we did not manage to have an impact

of damages on debt dynamics as might have been expected, as debt to GDP ratio had a negative peak instead of a positive peak. This may be due to the fact that this model treats spending as a constant exogenous share of GDP, rather than a constant level that may enable the ratio to move and be agnostic to GDP movements. We are also only considering a lump-sum tax that has no distorsive effect on real variables, which means that the fiscal rule can infinitely increase tax rates to compensate a rise in debt without hurting the economy. The introduction of distorsive taxes may solve this issue. Finally, we calibrated the our fiscal rule with the highest tax inertia compatible with a convergence of all specifications, which is probably not high enough to simulate a more realistic scenario where debt would be allowed to be able to increase. Under a small carbon budget those issues don't appear, since the implied ambitious mitigation requirements automatically entail fiscal stress (in a public mitigation scenario), but are unveiled when this budget relaxed and effort spread along a larger time horizon.

4.2 Policy implications

We thus found that in the case of ambitious mitigation objectives, compatible with global mean temperature staying under +1.5°C compared to pre-industrial times, relying solely on public money to finance the transition can prove risky, due to the asymmetry between public and private mitigation costs. A reasonable solution may be to find a middle ground and split private and public mitigation effort. Seghini and Dees (2025) showed indeed that such a path, with a public share rising from 0 to 40%, is the one that maximizes welfare in the economy. Furthermore, the physical damages produced by climate change, in addition to reducing GDP, exacerbate the negative effects of transition policies, with the peak default probability rising from about 12% to above 20% in the worst case scenario. The cost to output, alongside consumption and investment, rises greatly under 3°C, with losses of about a third of output under the most severe damage specifications.

It is important in interpreting the model's results to keep in mind that the the dynamics displayed and commented above are the dynamics around the long-term trend of GDP, aligned on the rate of growth of technology z, growth calibrated at 2%. We are thus reasoning in potential and not observed GDP. The 34% shock of climate damages in a 3°C world under the Bilal & Känzig specification does not mean that GDP will be decreased by 34% from what it is today. It means that it will be down by 34% from what it would have been if we stopped emitting CO2 today. Assuming that the economy grows on average at 2% per year over the next 125 years, a world stabilized at 3°C would be only 8 times richer than today instead of 12 times.

Those numbers have nonetheless to be put in perspective. One may wonder whether sustaining such a growth trend, purely driven by technology, is feasible, even with diffusion of new technologies like AI. Secondly, we also make the assumption in the

model that the emissions are to get decoupled from GDP at the same rate as this long-term growth, meaning that those technological advances stimulating the economy's growth potential are equally good for reducing emissions. One may mention the higher energy efficiency of new products, the advent of green technologies, enabling to shift away from fossil fuels to cleaner sources of energy. But the net impact on climate of the diffusion of AI, our "new industrial revolution", considering the amount of energy required to build and maintain the computing power necessary to address the demand for tokens, this net impact on climate may well be negative, not to say extremely negative. The economy may thus get much less than "8 times richer".

Furthermore, the damage function we use is only an approximation based on hypotheses and calibrations relying on what has been seen in the past. The dynamics produced by climate change, as temperatures go higher, may well be more pronounced than what we though they were. The damages described also don't include all the effects, such as the increased link with political instability, conflicts, war, that may indirectly result from climate change, that may not be captured in historical data.

Policymakers should thus strive to maintain global average temperatures to a reasonable level. Though 1.5°C degrees may be too ambitious, 3°C is nonetheless too much, and one could set an intermediary target at for instance 2°C (the upper bound of the Paris Agreement's objective). As mentioned earlier, such action should be taken balancing the impacts of transition policies with those of climate damages, and the solution probably lies in an optimal policy mix, combining both a carbon tax and public mitigation, as mentioned earlier. The model used in this work, as practical as it is to analyze transition scenarios, cannot be by definition fully realistic, take all dimensions of the debt-climate nexus at the same time. One may wonder for instance about the political feasibility of a carbon tax, or the redistribution effects of transition policies. Those axes are left for future research.

5 Discussion

The Seghini and Dees (2025) model presents a lot of qualities. It provides in a single framework many key mechanisms from IAMs and the debt sustainability literature, all plugged into a canonical RBC model. This enables us to analyze the interaction between climate change and debt sustainability while avoiding too much complexity (heterogeneous agents, monetary policy, stochasticity). It makes it at the same time polyvalent, easy to read, and comparable to other papers in the literature. However, one can always flip the coin and say that it presents the "limits of its qualities". Indeed, the perfect foresight algorithm implies that all information is transparent, available and forseable by the agents, who exhibit purely rational behavior, the latter being standard in modern macroeconomic models. Agent homogeneity, by entailing that households are all equally owners of firms' capital, or that climate damages and transition policies impact every one the same by may be reasonably put into question. Those criticisms may however be made of any model as it is impossible to include all aspects of a topic at a same time, one must chose his battles.

However, the version of the model used in this document could directly benefit from improvements not too hard to implement. One may want to introduce distorsive taxes so that raising them eventually presents some costs. One may also deem perfectly elastic government spending (implied by a fixed GDP ratio) as not realistic and model government introduce some inertia. One easy way to do that could be to fix government spending in levels so that it is totally inelastic to GDP, enabling the government spending to GDP ratio to move along the green transition. A reduction in GDP produced by extreme climate events would indeed not be plausibly followed by a reduction in public spending, quite the opposite. One may also want to directly model emergency spending resulting from climate events. More prosaically, we encountered some convergence issues under the exact calibration provided in Seghini and Dees (2025), notably because of the inertia of the fiscal stabilisation rule, being hard to sufficiently raise without loosing some scenarios. Adapting this rule and the way it is calibrated may enable us to raise this inertia and reach more economically sensible results for the impact of climate change on debt to GDP ratios and the probability of government default. Finally, the model equations, as currently implemented, assume that the government's fiscal balance is systematically positive, which is not consistent with the fiscal reality of France. One may want to force a deficit at the beginning of the simulations. From a more general perspective, the modelling of the climate-debt nexus may also benefit from empirical analysis. Some of the calibrations, such as the debt limit, that is used to calculate the probability of sovereign default, may need to be re-estimated taking into account the impact of climate damages. Finally, a bridge could be made between the model's framework and more "down-to-earth" approches

as the DSA methodologies used by international institutions and the ECB to evaluate whether they should lend money to governments.

6 Conclusion

The aim of this work was to study the relationship between public debt sustainability and climate change. We take the case of France, where, after decades of accumulated deficits resulting in rising debt, questions on its sustainability get back on stage as the perspective of a return to an output growth rate higher than the interest rate paid on debt seems ever less plausible. We chose to carry out this analysis within the framework of Seghini and Dees (2025), a dynamic general equilibrium (DGE) model that includes the green transition, climate damages and a fiscal block a dynamic general equilibrium (DGE) model, in order to simulate the economic dynamics resulting from the shock represented by the net-zero transition.

We managed to replicate its main results, isolating the pure effect of transition policies from that of physical damages. In the case of ambitious mitigation objectives, compatible with global mean temperature staying under +1.5°C compared to pre-industrial times, relying solely on public money to finance the transition can prove risky due to the difference between public and private mitigation cost structures. The physical damages produced by climate change, in addition to reducing GDP, exacerbate the fiscal stress that may arise from government-funded transition policies. The solution may be to find the optimal split between private and public mitigation effort. Seghini and Dees (2025) showed that a public share rising from 0 to 40% between today and the end of the transition is the one that maximizes welfare in the economy.

We then introduce a new damage function, taken from Bilal and Känzig (2026), on the upper end of the damage spectrum found in the literature. The authors had shaken the literature consensus by showing that, as a result of taking world temperatures instead of country temperature, the impact of climate change on the economy was 10 times as strong as what was previously agreed upon. The GDP loss under a +3°C compared to pre-industrial times would entail a 53% loss in GDP with respect to a no-damage baseline. Their damage function models the dynamic impact of temperature, as opposed to the static impact of those commonly found in the literature. We find, after plugging it into the Seghini and Dees (2025) model and comparing with a range of other damage specifications, that it is indeed the most severe in all our scenarios, although the recalibrated specification of Seghini et al. (2026) comes very close in the 3°C world. Its effect on output is about 14 times that of the latest DICE specification, and reaches a loss of about a third of GDP under a +3°C scenario.

Although the model is a simplification of reality, it enables us to capture the key dynamics involving public debt and the impact of climate change on the economy, both through transition and policies and physical damages. This approach may nonetheless still be extended and complemented with empirical analyses. This subject being still understudied at the moment, a significant gap is left for future research. I was given

the opportunity to extend this work at the Bank of France alongside Caterina Seghini, where I am currently developing the Seghini et al. (2026) model and exploring bridges between with the institution's debt sustainability analysis framework.

References

Agarwala, M., Burke, M., Klusak, P., Mohaddes, K., Volz, U., and Zenghelis, D. (2021). Climate change and fiscal sustainability: Risks and opportunities. *National Institute Economic Review*, 258(1):28–46.

Annicchiarico, B. and Di Dio, F. (2015). Environmental policy and macroeconomic dynamics in a new Keynesian model. *Journal of Environmental Economics and Management*, 69:1–21.

Barrage, L. (2020a). The fiscal costs of climate change. *AEA Papers and Proceedings*, 110:107–112.

Barrage, L. (2020b). Optimal dynamic carbon taxes in a climate-economy model with distortionary fiscal policy. *Review of Economic Studies*, 87(1):1–39.

Barrage, L. and Nordhaus, W. (2024). Policies, projections, and the social cost of carbon: Results from the DICE-2023 model. *Proceedings of the National Academy of Sciences*, 121(13):e2312030121.

Battiston, S. and Monasterolo, I. (2020). The climate spread of corporate and sovereign bonds. Working Paper 3376218, SSRN.

Bilal, A. and Känzig, D. R. (2026). The macroeconomic impact of climate change: Global versus local temperature. *Quarterly Journal of Economics*, 141(2):889–944.

Blanchard, O. (2019). Public debt and low interest rates. *American Economic Review*, 109(4):1197–1229.

Blanchard, O. J. and Leigh, D. (2014). Learning about fiscal multipliers from growth forecast errors. *IMF Economic Review*, 62(2):179–212.

Burke, M., Hsiang, S. M., and Miguel, E. (2015). Global non-linear effect of temperature on economic production. *Nature*, 527:235–239.

Calvo, G. A. (1988). Servicing the public debt: The role of expectations. *American Economic Review*, 78(4):647–661.

Cohen, D. and Sachs, J. (1986). Growth and external debt under risk of debt repudiation. *European Economic Review*, 30(3):529–560.

Collard, F., Habib, M., and Rochet, J.-C. (2015). Sovereign debt sustainability in advanced economies. *Journal of the European Economic Association*, 13(3):381–420.

Corsetti, G., Kuester, K., Meier, A., and Müller, G. J. (2013). Sovereign risk, fiscal policy, and macroeconomic stability. *Economic Journal*, 123(566):F99–F132.

Dell, M., Jones, B. F., and Olken, B. A. (2012). Temperature shocks and economic growth: Evidence from the last half century. *American Economic Journal: Macroeconomics*, 4(3):66–95.

Dietz, S. and Venmans, F. (2019). Cumulative carbon emissions and economic policy: In search of general principles. *Journal of Environmental Economics and Management*, 96:108–129.

Garcia-Macia, D., Lam, W. R., and Nguyen, A. D. M. (2024). Public debt dynamics during the climate transition. IMF Working Paper WP/24/71, International Monetary Fund.

Ghosh, A. R., Kim, J. I., Mendoza, E. G., Ostry, J. D., and Qureshi, M. S. (2013). Fiscal fatigue, fiscal space and debt sustainability in advanced economies. *Economic Journal*, 123(566):F4–F30.

Heutel, G. (2012). How should environmental policy respond to business cycles? Optimal policy under persistent productivity shocks. *Review of Economic Dynamics*, 15(2):244–264.

IPCC (2023). Climate change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Technical report, Intergovernmental Panel on Climate Change, Geneva. Core Writing Team, H. Lee and J. Romero (eds.).

Kalkuhl, M. and Wenz, L. (2020). The impact of climate conditions on economic production. Evidence from a global panel of regions. *Journal of Environmental Economics and Management*, 103:102360.

Kaya, Y. (1990). Impact of carbon dioxide emission control on GNP growth: Interpretation of proposed scenarios. Paper presented to the IPCC Energy and Industry Subgroup, Response Strategies Working Group, Paris (mimeo).

Klusak, P., Agarwala, M., Burke, M., Kraemer, M., and Mohaddes, K. (2023). Rising temperatures, falling ratings: The effect of climate change on sovereign creditworthiness. *Management Science*, 69(12):7468–7491.

Krugman, P. (1988). Financing vs. forgiving a debt overhang. *Journal of Development Economics*, 29(3):253–268.

Krugman, P. (2011). The bleeding cure. The New York Times. Opinion, 19 September 2011.

Matthews, H. D., Gillett, N. P., Stott, P. A., and Zickfeld, K. (2009). The proportionality of global warming to cumulative carbon emissions. *Nature*, 459:829–832.

Meadows, D. H., Meadows, D. L., Randers, J., and Behrens III, W. W. (1972). *The Limits to Growth: A Report for the Club of Rome's Project on the Predicament of Mankind*. Universe Books, New York.

Nordhaus, W. D. (1992). An optimal transition path for controlling greenhouse gases. *Science*, 258(5086):1315–1319.

Pisani-Ferry, J. and Mahfouz, S. (2023). Les incidences économiques de l'action pour le climat. Rapport à la Première ministre, France Stratégie, Paris.

Reinhart, C. M. and Rogoff, K. S. (2010). Growth in a time of debt. *American Economic Review*, 100(2):573–578.

Reinhart, C. M. and Rogoff, K. S. (2013). Reinhart and Rogoff: Responding to our critics. The New York Times. Opinion, 26 April 2013.

Seghini, C. (2024). Climate and sovereign debt sustainability: A literature review. SSRN Working Paper 4766658, Swiss Finance Institute and Université de Genève.

Seghini, C. and Dees, S. (2025). The green transition and public finances. *Macroeconomic Dynamics*, 29:e152.

Seghini, C., Dees, S., and De Gaye, A. (2026). Under pressure: Sovereign debt challenges in a warming world. Unpublished working paper, Banque de France.

Stern, N. (2006). Stern Review: The economics of climate change. Technical report, HM Treasury, London. Published in book form by Cambridge University Press, 2007.

Strassmann, K. M. and Joos, F. (2018). The Bern Simple Climate Model (BernSCM) v1.0: An extensible and fully documented open-source re-implementation of the Bern reduced-form model for global carbon cycle–climate simulations. *Geoscientific Model Development*, 11:1887–1908.

Weitzman, M. L. (2010). What is the "damages function" for global warming – and what difference might it make? *Climate Change Economics*, 1(1):57–69.

Zenios, S. A. (2022). The risks from climate change to sovereign debt. *Climatic Change*, 172:30.

List of Figures

1 Global warming and recorded natural disasters 1
2 Public debt, fiscal balance and interest-growth differential in France, Germany and Italy, 1980–2025 2
3 The five damage specifications as a function of warming 21
4 Macro-financial effects of the five policy scenarios, 1.5 °C world, no climate damage 25
5 Annual default probability under the five policy scenarios, 1.5 °C world, no climate damage 26
6 The five policy scenarios in the 1.5 °C and 3 °C worlds, no climate damage 27
7 GDP, public debt and default probability under the five damage specifications, 1.5 °C world 28
8 GDP under the five damage specifications, carbon tax scenario, 3 °C world 29

Appendices

These appendices gather the material that supports Section 3 but is not needed to follow the argument. Appendix A derives the recursive form of the damage function of Bilal and Känzig (2026) that we use in the simulations. Appendix B gives the detrended equations of the model and the two steady states between which the transition takes place.

A Mathematical derivation of the Bilal & Känzig damage function

Section 3.4.3 states the two-state form of the damage function without proof. It rests on three steps: splitting the convolution integral into two exponentially weighted averages of past warming, showing that each average follows a simple differential equation, and solving that equation over one quarter. The last part of this appendix looks at the long run of the two states, which gives the static version of the function drawn in Figure 3, and at the way the function is normalised in the simulations.

Throughout, $\widehat{T}_t$ is warming above pre-industrial levels, s is the time elapsed since a temperature shock, measured in years, and $\zeta_s = A\left(e^{-Bs} - e^{-Cs}\right)$ is the kernel, with $C > B > 0$. As in Section 3.4.2, $D_{f,t}$ denotes the damage factor that multiplies total factor productivity, written Z_t in Bilal and Känzig (2026).

A.1 From the convolution integral to two differential equations

Splitting the integral. Starting from the convolution integral of Bilal and Känzig (2026),

$$\log D_{f,t} = \int_0^\infty \zeta_s \, \widehat{T}_{t-s} \, ds = \int_0^\infty A\left(e^{-Bs} - e^{-Cs}\right) \widehat{T}_{t-s} \, ds, \tag{A1}$$

the linearity of the integral lets us write it as the difference of two terms:

$$\log D_{f,t} = A \underbrace{\int_0^\infty e^{-Bs} \, \widehat{T}_{t-s} \, ds}_{X_t^B} - A \underbrace{\int_0^\infty e^{-Cs} \, \widehat{T}_{t-s} \, ds}_{X_t^C} = A\left(X_t^B - X_t^C\right). \tag{A2}$$

Each state is a weighted sum of past warming, in which a shock received s years ago is discounted at rate B or C. Since $B < C$, X_t^B keeps a long memory of past warming and X_t^C a short one.

Change of variable. We derive the law of motion of X_t^B; the one of X_t^C follows by replacing B with C. Setting $u = t - s$, so that $du = -ds$, the lower bound $s = 0$

becomes $u = t$ and the upper bound $s \to +\infty$ becomes $u \to -\infty$:

$$X_t^B = \int_t^{-\infty} e^{-B(t-u)}\, \widehat{T}_u\, (-du) = \int_{-\infty}^{t} e^{-B(t-u)}\, \widehat{T}_u\, du. \tag{A3}$$

Reversing the bounds absorbs the minus sign. The variable t now appears in two places: in the upper bound, and inside the integral through $e^{-B(t-u)}$.

Differentiating with respect to t. Because t enters both the bound and the integrand, we need the general form of the Leibniz integral rule, for bounds that depend on the variable:

$$\frac{d}{dx}\int_{a(x)}^{b(x)} f(x,u)\, du = f\big(x,b(x)\big)\, b'(x) - f\big(x,a(x)\big)\, a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f(x,u)}{\partial x}\, du. \tag{A4}$$

Here $x = t$, $a(t) = -\infty$ so that $a'(t) = 0$ and the lower-bound term drops, $b(t) = t$ so that $b'(t) = 1$, and $f(t,u) = e^{-B(t-u)}\, \widehat{T}_u$. The upper-bound term is $f(t,t) = \widehat{T}_t$, and the integral term is

$$\int_{-\infty}^{t} \frac{\partial}{\partial t}\left[e^{-B(t-u)}\right] \widehat{T}_u\, du = -B \int_{-\infty}^{t} e^{-B(t-u)}\, \widehat{T}_u\, du = -B\, X_t^B.$$

Adding the two terms, and doing the same for X_t^C, gives

$$\dot{X}_t^B = \widehat{T}_t - B\, X_t^B, \qquad \dot{X}_t^C = \widehat{T}_t - C\, X_t^C. \tag{A5}$$

Each state is fed by current warming and forgets its past at its own rate. This is how Strassmann and Joos (2018) turn the impulse response of atmospheric carbon into a system of separate boxes: each exponential term of the response becomes one box, which receives part of the input and has its own turnover time.

A.2 Exact discretisation at a quarterly frequency

The model runs at a quarterly frequency while B and C are annual rates, so one period lasts $\Delta t = 0.25$ year. We solve (A5) over one period, holding warming constant within the period at a value $\widehat{T}$. Written as $\dot{x} + B\, x = \widehat{T}$, the equation is linear with a constant coefficient, and e^{Bs} is an integrating factor, since

$$\frac{d}{ds}\Big(e^{Bs}\, x(s)\Big) = e^{Bs}\, (\dot{x}(s) + B\, x(s)) = e^{Bs}\, \widehat{T}.$$

Integrating both sides from $s = t - \Delta t$ to $s = t$ gives

$$e^{Bt}\, x(t) - e^{B(t-\Delta t)}\, x(t-\Delta t) = \widehat{T} \int_{t-\Delta t}^{t} e^{Bs}\, ds = \widehat{T}\, \frac{e^{Bt} - e^{B(t-\Delta t)}}{B},$$

and dividing by e^{Bt},

$$X_t^B = e^{-B\Delta t}\, X_{t-\Delta t}^B + \widehat{T}\,\frac{1-e^{-B\Delta t}}{B}, \qquad X_t^C = e^{-C\Delta t}\, X_{t-\Delta t}^C + \widehat{T}\,\frac{1-e^{-C\Delta t}}{C}. \tag{A6}$$

The first term is the memory inherited from the previous quarter, which fades at rate B. The second is the warming of the quarter, which adds to that memory. Its weight $(1-e^{-B\Delta t})/B$ is slightly smaller than Δt, because the warming received early in the quarter has already started to fade by the end of it. When $B\Delta t$ is small, this weight tends to Δt and $e^{-B\Delta t}$ to $1-B\Delta t$, which gives back the usual Euler scheme, $X_t^B = (1-B\Delta t)\, X_{t-\Delta t}^B + \Delta t\, \widehat{T}$. Unlike the Euler scheme, (A6) adds no approximation of its own: once warming is taken as constant within the quarter, it is the exact solution of (A5), whatever the size of $B\Delta t$. Strassmann and Joos discretise their carbon cycle in the same way (their Appendix B).

In the simulations, time is counted in quarters and the warming held constant over quarter t is the one reached in the previous quarter, $\widehat{T}_{t-1}$:

$$X_t^B = e^{-B/4}\, X_{t-1}^B + \widehat{T}_{t-1}\,\frac{1-e^{-B/4}}{B}, \qquad X_t^C = e^{-C/4}\, X_{t-1}^C + \widehat{T}_{t-1}\,\frac{1-e^{-C/4}}{C}. \tag{A7}$$

Both states are then known at the start of the quarter, so that productivity does not react within the quarter to the emissions produced during it.

A.3 Long run, initial condition and normalisation

Long run. When warming stays at a constant level $\widehat{T}^\star$, the two states of (A7) settle at a fixed point. Setting $X_t^B = X_{t-1}^B = X^{B\star}$ gives $X^{B\star}\left(1-e^{-B/4}\right) = \widehat{T}^\star\left(1-e^{-B/4}\right)/B$, hence

$$X^{B\star} = \frac{\widehat{T}^\star}{B}, \qquad X^{C\star} = \frac{\widehat{T}^\star}{C}. \tag{A8}$$

These are exactly the values at which $\dot{X}^B$ and $\dot{X}^C$ vanish in (A5): the discretisation does not move the long run. Substituting them in (A2),

$$\log D_f^\star = A\left(\frac{1}{B}-\frac{1}{C}\right)\widehat{T}^\star = \widehat{T}^\star \int_0^\infty \zeta_s\, ds, \tag{A9}$$

since $\int_0^\infty \left(e^{-Bs}-e^{-Cs}\right) ds = 1/B - 1/C$. In the long run, the two-kernel function therefore reduces to an exponential in warming, with slope $A(1/B-1/C)$. This is the version drawn in Figure 3 and written D_5 in Section 3.4.3. Two properties of the kernel explain the dynamics seen before this long run is reached. A shock has no effect on impact, since $\zeta_0 = A(1-1) = 0$. Its effect then peaks when $B\,e^{-Bs} = C\,e^{-Cs}$, that is $s^\star = \ln(C/B)/(C-B)$ years after the shock, and fades afterwards.

Initial condition. The simulation does not start at the beginning of warming: at the start date, the world is already $\widehat{T}_0$ above pre-industrial levels (Table 3). We therefore start both states at the values that this warming would have produced if it had been constant, $X^B = \widehat{T}_0/B$ and $X^C = \widehat{T}_0/C$. This amounts to assuming that past warming was slow compared with the memory of the kernel, which is a reasonable approximation after a century of gradual warming.

Normalisation. As for the other damage specifications, the damage factor is normalised to its value at the start date, so that the initial economy is the same under every specification and only the warming that occurs after the start date lowers productivity:

$$D_{f,t} = \bar{D} \exp\left\{ A\left(X_t^B - X_t^C\right) - A\left(\frac{1}{B} - \frac{1}{C}\right)\widehat{T}_0 \right\}. \tag{A10}$$

At the start date, the exponent is zero and $D_f = \bar{D}$. Once warming has settled at $\widehat{T}^\star$, $D_f = \bar{D}\exp\left\{A\,(1/B - 1/C)\left(\widehat{T}^\star - \widehat{T}_0\right)\right\}$.

B Detrended and steady-state equations

The equations of Section 3.2 are written in levels, and two trends run through them: labour efficiency $Z_t = e^{zt}$ grows at rate z, and the carbon intensity of output $\xi_t = \xi_0\, e^{-\omega t}$ falls at rate ω. Output and most quantities therefore grow without bound, and the system has no steady state in levels. As explained in Section 3.3.1, we remove these trends before solving the model. This appendix lists the transformed variables, the detrended equations and the two steady states.

B.1 Detrending

Variables that grow with labour efficiency are divided by e^{zt} and written in lower case. Marginal utilities fall as consumption grows, so they are multiplied by e^{zt} instead. The carbon tax per tonne has to rise as carbon intensity falls, so it is multiplied by $e^{-\omega t}$. Emissions grow at rate $z - \omega$, which is zero in our calibration since $\omega = z$ (Table 1), so emissions and cumulative emissions need no transformation. Shares, rates, ratios and hours are stationary and are left as they are.

Once the trend is removed, marginal utility falls by the factor e^{-z} from one quarter to the next, so the discount factor becomes $\bar{\beta} = \beta/e^z$.

Table 7: Detrended variables

Variable in levels	Detrended variable	Reason
Y_t, C_t, I_t, W_t	$y_t = Y_t / e^{zt}$, and likewise c_t, i_t, w_t	grow with labour efficiency
K_{t+1}	$k_t = K_{t+1} / e^{z(t+1)}$	chosen in t, used in production in $t+1$
λ_t, λ_t^k	$\lambda_t^z = \lambda_t e^{zt}$, and $\lambda_t^k e^{zt}$, still written λ_t^k	fall as consumption grows
τ_t^E	$P_t^E = \tau_t^E e^{-\omega t}$	rises as carbon intensity falls
E_t, H_t, H_t^G	unchanged	trend $z - \omega$ equal to zero
L_t, m_t, m_t^g, Θ_t, τ_t, s_t, b_t, PD_t, R_t, R_t^d, R_t^k, q_t^k	unchanged	stationary

B.2 Detrended equations

Under perfect foresight, agents know the future values of all variables, so the expectation operators of Section 3.2 can be dropped.

Environment. Since $z = \omega$, the environment block contains no trending variable, and (E1) to (E4) keep their form. Warming follows cumulative world emissions through the TCRE (Section 3.4.1):

$$\widehat{T}_t = \widehat{T}_0 + \lambda_{\mathrm{TCRE}} \left(H_t^G - H_{-1}^G \right), \tag{B1}$$

which gives back $\widehat{T}_t \approx \lambda_{\mathrm{TCRE}} H_t^G$, because the initial warming $\widehat{T}_0$ equals $\lambda_{\mathrm{TCRE}} H_{-1}^G$ (Table 3).

Firms. Dividing the production function (F1) by e^{zt}, with $K_t = e^{zt} k_{t-1}$, gives

$$y_t = D_{f,t} \, k_{t-1}^{\kappa} \, L_t^{1-\kappa}. \tag{B2}$$

Emissions (F2) are $E_t = \xi_0 \, e^{(z-\omega)t} \, (1 - m_t - m_t^g) \, y_t$, that is

$$E_t = \xi_0 \, (1 - m_t - m_t^g) \, y_t. \tag{B3}$$

The carbon tax enters the costs of firms through $\tau_t^E \, \xi_t = P_t^E \, \xi_0$. The wedge (F6) and the first-order conditions (F7) to (F9) become

$$\Theta_t = 1 - (1 - s_t^A) \, \theta_1 \, m_t^{\theta_2} - P_t^E \, \xi_0 \, (1 - m_t - m_t^g), \tag{B4}$$

$$w_t = (1 - \kappa) \, \Theta_t \, \frac{y_t}{L_t}, \qquad R_t^k = \kappa \, \Theta_t \, \frac{y_t}{k_{t-1}}, \qquad P_t^E = \frac{(1 - s_t^A) \, \theta_1 \theta_2}{\xi_0} \, m_t^{\theta_2 - 1}. \tag{B5}$$

The abatement costs (F3) and (F4) are proportional to output and simply carry y_t instead of Y_t.

Households. The marginal utility of consumption (H7) and labour supply (H8) become

$$\lambda_t^z = \frac{1}{c_t}, \qquad \mu_L \, L_t^{\sigma_l} = \lambda_t^z \, w_t. \tag{B6}$$

Investment growth is $x_t = I_t / I_{t-1} = e^z \, i_t / i_{t-1}$, so that the adjustment cost $S(x_t)$ and its derivative are zero along the trend, where $x_t = e^z$. Capital accumulation (H4) becomes

$$k_t = \frac{1-\delta}{e^z} \, k_{t-1} + \frac{i_t}{e^z} \left[1 - S(x_t) \right], \tag{B7}$$

and the first-order conditions on investment (H11) and capital (H10), the latter with the spillover term of (H13), become

$$\lambda_t^z = \lambda_t^k \left[1 - S(x_t) - S'(x_t) \, x_t \right] + \bar{\beta} \, \lambda_{t+1}^k \, S'(x_{t+1}) \, x_{t+1}^2, \tag{B8}$$

$$\lambda_t^k = \bar{\beta} \, \lambda_{t+1}^z \, R_{t+1}^k + \bar{\beta} \, (1-\delta) \, \lambda_{t+1}^k \left(1 - \omega_q \, f \bar{\vartheta} \, \mathrm{PD}_{t+1} \right), \tag{B9}$$

where $\omega_q = 0$ outside the spillover simulations (Table 5). Tobin's Q (H12) and the risk-free rate (H14) are

$$q_t^k = \frac{\lambda_t^k}{\lambda_t^z}, \qquad R_t = \frac{\lambda_t^z}{\bar{\beta} \, \lambda_{t+1}^z}. \tag{B10}$$

The bond first-order condition (H9) does not need to be written separately: together with (B10), it gives the sovereign spread (H15), which the model imposes with equality, $R_t^d = R_t / (1 - f \bar{\vartheta} \, \mathrm{PD}_{t+1})$.

Government. Dividing the debt accumulation (G1) by output, with $Y_{t-1}/Y_t = y_{t-1}/(e^z \, y_t)$, gives

$$b_t = \frac{R_{t-1}^d \, y_{t-1}}{e^z \, y_t} \, b_{t-1} - s_t, \tag{B11}$$

which is (G4) with $1 + g_t^Y = e^z \, y_t / y_{t-1}$. The primary surplus (G2), as a share of output, uses the detrended carbon price:

$$s_t = \tau_t + (1 - \omega_g) \, P_t^E \, \xi_0 \, (1 - m_t - m_t^g) - g - s_t^A \, \theta_1 \, m_t^{\theta_2} - \theta_3 \, (m_t^g)^{\theta_4}. \tag{B12}$$

The default condition (G5), the probability of default (G6) and the fiscal rule (G7) and (G8) only involve ratios, rates and probabilities, and keep their form.

Market clearing. Public consumption includes the recycled share of carbon revenue. The resource cost of default G_t^d is zero along the simulated paths, where default only matters through its probability. Dividing (M) by e^{zt} then gives

$$y_t = c_t + i_t + \left[g + \omega_g \, P_t^E \, \xi_0 \, (1 - m_t - m_t^g) \right] y_t + \theta_1 \, m_t^{\theta_2} \, y_t + \theta_3 \, (m_t^g)^{\theta_4} \, y_t. \tag{B13}$$

B.3 The two steady states

The transition starts from a brown economy, which still emits, and ends in a green economy, where the carbon budget is exhausted and emissions have stopped. As discussed in Section 3.3.1, only the green economy is a true steady state: the brown economy keeps adding to its stock of emissions, so its values serve as initial conditions. Both are obtained from the equations of Section B.2 with all variables held constant, investment growth at $x^\star = e^z$ and debt at its target $b^\star$.

Rates and sovereign risk. Equations (B8) to (B10) and (H15) give $q^{k\star} = 1$, $\lambda^{k\star} = \lambda^{z\star}$ and

$$\begin{aligned} R^\star &= \frac{e^z}{\beta}, \qquad \mathrm{PD}^\star = \Phi\left(\frac{\ln b^\star - \ln b^M}{\sigma_B}\right), \\ R^{d\star} &= \frac{R^\star}{1 - f\bar{\vartheta}\,\mathrm{PD}^\star}, \qquad R^{k\star} = \frac{e^z}{\beta} - (1-\delta)\left(1 - \omega_q\, f\bar{\vartheta}\,\mathrm{PD}^\star\right). \end{aligned} \tag{B14}$$

These values are the same in both states, since debt is at its target in both.

Carbon budget, abatement and warming. The two states differ by how much of the carbon budget has been spent. In the brown economy, no policy instrument is in force yet ($\theta = s^A = \omega_g = 0$), the national stock is at its initial value and warming at its start-date level:

$$H^\star = H_{-1}, \qquad m^\star + m^{g\star} = \left(\frac{H_{-1}}{\bar{E}_0^n}\right)^p, \qquad m^{g\star} = 0, \qquad \widehat{T}^\star = \widehat{T}_0. \tag{B15}$$

In the green economy, the national budget is exhausted, abatement is complete and emissions are zero. The instruments take the values of the scenario (Table 6), and the world has spent its own budget:

$$H^\star = \bar{E}_0^n, \qquad m^\star + m^{g\star} = 1, \qquad m^{g\star} = \theta, \qquad \widehat{T}^\star \approx \widehat{T}_0 + \lambda_{\mathrm{TCRE}}\,\bar{E}_0^G. \tag{B16}$$

Warming therefore settles at about 1.53°C in the 1.5°C world and 3.02°C in the 3°C world. The carbon price and the wedge follow from (B4) and (B5). In the green economy, the carbon term of (B4) vanishes and $\Theta^\star = 1 - (1 - s^A)\,\theta_1\,(m^\star)^{\theta_2}$. The damage factor is $D_f^\star = \bar{D}$ in the brown economy, by the normalisation of Section A.3, and is evaluated at $\widehat{T}^\star$ in the green economy.

Ratios and levels. The capital demand in (B5) and capital accumulation (B7) give

$$\frac{k^\star}{y^\star} = \frac{\kappa\,\Theta^\star}{R^{k\star}}, \qquad \frac{i^\star}{y^\star} = (e^z - 1 + \delta)\,\frac{k^\star}{y^\star}, \tag{B17}$$

and market clearing (B13) gives the consumption share:

$$\frac{c^\star}{y^\star} = 1 - \frac{i^\star}{y^\star} - g - \omega_g\, P^{E\star}\,\xi_0\,(1 - m^\star - m^{g\star}) - \theta_1\,(m^\star)^{\theta_2} - \theta_3\,(m^{g\star})^{\theta_4}. \tag{B18}$$

Combining labour supply (B6) with labour demand (B5) gives $\mu_L\,(L^\star)^{1+\sigma_l} = (1-\kappa)\,\Theta^\star\,y^\star/c^\star$, and the production function (B2) with $k^\star = (k^\star/y^\star)\,y^\star$ then gives output:

$$L^\star = \left[\frac{(1-\kappa)\,\Theta^\star}{\mu_L\,c^\star/y^\star}\right]^{\frac{1}{1+\sigma_l}}, \qquad y^\star = \left[D_f^\star\left(\frac{k^\star}{y^\star}\right)^{\kappa}\right]^{\frac{1}{1-\kappa}} L^\star. \tag{B19}$$

All other levels follow from these ratios.

Fiscal variables. With debt at its target, (B11) gives the primary surplus that keeps it there, and (B12) the tax rate that delivers this surplus:

$$\begin{aligned} s^\star &= b^\star\left(\frac{R^{d\star}}{e^z}-1\right), \\ \tau^\star &= s^\star + g + s^A\,\theta_1\,(m^\star)^{\theta_2} + \theta_3\,(m^{g\star})^{\theta_4} - (1-\omega_g)\,P^{E\star}\,\xi_0\,(1-m^\star-m^{g\star}). \end{aligned} \tag{B20}$$

In the green economy, the last term is zero since nothing is left to tax, and the tax rate has to cover, permanently, the subsidies and the public abatement of the scenario. This terminal value is the $\tau^\star$ towards which the fiscal rule (G7) steers taxation, and it is higher in the scenarios where the state subsidises or performs abatement.